

Microwave as a Tool for Synthesis of Carbon-Based Electrodes for Energy Storage

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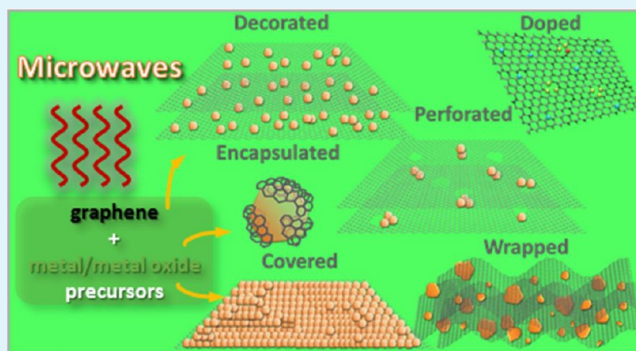
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ABSTRACT: This Spotlight on Applications highlights the significant impact of microwave-assisted methods for synthesis and modification of carbon materials with enhanced properties for electrodes in energy storage applications (supercapacitors and batteries). For the past few years, microwave irradiation has been increasingly used for the synthesis of carbon materials with different morphologies using various precursors. Microwave processing exhibits numerous advantages, such as short processing times, high yield, expanded reaction conditions, high reproducibility, and high purity of products. On this frontier research area, we have discussed microwave-assisted synthesis, defect creation, simultaneous reduction and exfoliation, and heteroatom doping in carbon materials. By careful manipulation of microwave irradiation parameters, the method becomes a powerful and efficient tool to generate different morphologies in carbon-based materials. Other important outcomes are the flexible control over the degree of reduction and exfoliation of graphene derivatives, the generation of defects in graphene-based materials by metals, the intercalation of metal oxides into graphene derivatives, and heteroatom doping of graphene materials. The Spotlight on Applications aims to provide a condensed overview of the current progress in carbon-based electrodes synthesized by microwave, pointing out outstanding challenges and offering a few suggestions to trigger more research endeavors in this important field.

KEYWORDS: carbon materials, microwave synthesis, graphene, defects, doping, electrodes, energy storage



1. INTRODUCTION

In the last few decades, carbon based materials and composites have attracted enormous attention for application in electrodes of energy storage devices.^{1–3} Exceptional properties, such as high conductivity, low density, high chemical and thermal stabilities, high surface area, high earth abundance, low cost, and environmentally-friendly nature, are some of the main reasons for carbons to be the most researched (and used) type of electrode material for electrochemical devices.⁴ Additionally, the physical and chemical properties of carbon-based electrodes, such as 3D network structure, degree of graphitization, mesoporous structure, surface area, functional groups, and doping content can all be manipulated in order to maximize the electrochemical performance according to the intended application.⁵ High surface area carbon derivatives provide good support for metal oxides, further improving their performance as electrode materials in energy storage applications.⁶

Carbon is an element with a very rich chemistry, enabling the obtention of a large variety of structures having different dimensions and morphologies.⁷ It has been observed that the appropriate dimensional match between the pore sizes of carbon

electrodes and the ion sizes of the electrolytes is a key factor for improving energy storage performance.^{8,9} The degree of graphitization of carbon materials is also very important, since high graphitization promotes electron transport. On the other hand, the formation of structural defects by doping carbon materials in supercapacitor (SC) electrodes normally leads to increased capacitance when compared to the pristine material.¹⁰

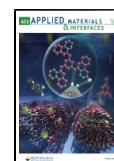
A variety of strategies have been proposed for tailoring graphene and its derivatives for energy storage applications, but the cost-effective processing of carbon electrodes for energy storage systems is still a big challenge.¹¹ In this context, the synthesis and modification of carbon materials by microwave (MW) irradiation has attracted much interest in recent times, due to the low initial investment and processing costs associated with MW methods.¹² In operational terms, the main advantage

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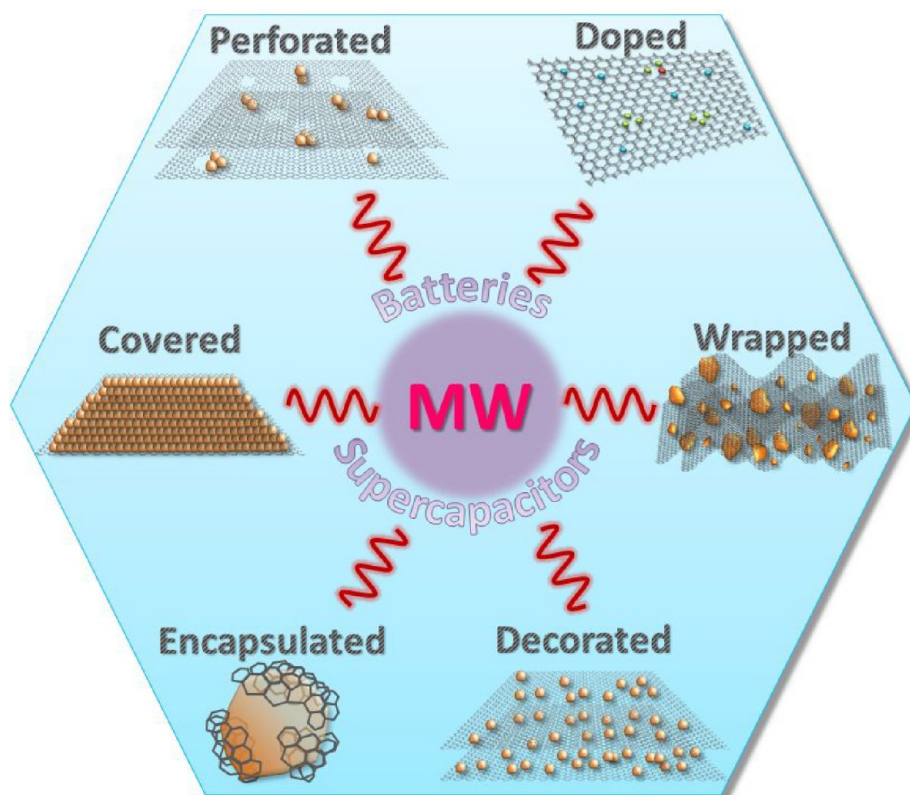


Figure 1. Schematic illustration showing the different alternative architectures for carbon-based composites synthesized by MW irradiation for SC and battery electrodes.

of MW methods is derived from the efficient energy transfer. Depending on the degree of absorption, the MW irradiation can promote fast volumetric heating of materials, decreasing the reaction times.¹³ When compared to conventional methods, other advantages are the noncontact heating, the low energy consumption, the possibility of applying selective heating, and the higher yields.¹⁴

It is well-known that most carbon materials are very good MW absorbers.¹⁵ As a consequence, the efficient heat energy input can be exploited for promoting several types of chemical reactions, including synthesis, reduction, exfoliation, doping, decoration, etc.¹⁶ Novel carbon-based composite electrodes combining metals/metal oxides with graphene derivatives, like graphene oxide (GO), reduced graphene oxide (rGO), graphene nanosheets (GNSs), and graphene nanoribbons (GNRs), have been researched for improved electrochemical performance. With increased electron transport, high surface area, and efficient electrolyte pathways, these hybrid materials have been used in SCs and batteries, helping to achieve high energy storage with ultralong cyclic stability.^{17–19} By adopting the MW synthesis approach, graphene-based composites containing metal oxides have been synthesized, exhibiting different structural configurations which could easily be achieved through small variations in MW power or irradiation time. The resulting graphene-based electrodes could be doped, perforated, and the metal/metal oxides could be anchored, intercalated/wrapped, or encapsulated by the graphene layers. Figure 1 shows a schematic illustration of different configurations achieved in carbon-based composites synthesized by MW irradiation for energy storage applications.

In this Spotlight on Applications, we have summarized and discussed recent developments on the MW-assisted fabrication

of graphene-based electrode materials for energy storage applications. After a brief introduction and a discussion about the advantages of MW heating, the synthesis and modification of carbon materials by MW-assisted processing is outlined. Heteroatom doping, defect generation in the carbon network, and hybrid materials for charge storage electrodes are then discussed, with the main focus being placed on the work from our research group. The text is finished with a discussion of the remaining challenges for processing carbon-based materials by MW irradiation.

2. MICROWAVE: A TOOL FOR DESIGNING CARBON-BASED ELECTRODES

MWs are nonionizing electromagnetic radiation with frequencies ranging from 0.3 to 300 GHz (corresponding to wavelengths between 1 m and 1 mm). In the electromagnetic spectrum, the MWs lie in the region between infrared radiation and radio waves, and for practical applications in radar and telecommunications, they are divided into three bands: ultrahigh frequency (UHF, 0.3 GHz–3 GHz), superhigh frequency (SHF, 3 GHz–30 GHz), and extremely high frequency (EHF, 30 GHz–300 GHz).²⁰ In order to avoid electromagnetic interference, especially in telecommunications, it is common practice for national governments to allocate certain frequencies for MW heating in industrial, research, medical, and domestic equipment. Currently, 2.45 GHz (wavelength, 12.2 cm) is by far the most commonly used frequency for domestic MW ovens; however, large industrial/commercial ovens often use 0.915 GHz (wavelength, 32.8 cm) due to the lower energy consumption. MW energy is derived from electrical energy with a conversion efficiency of

approximately 50% at 2.45 GHz, while at 0.915 GHz the efficiency increases to 85%.^{21,22}

2.1. Conventional Heating and Microwave Heating. In conventional thermal treatments, heat is transferred from the surface toward the center of the sample by conduction, convection (for fluids), and to a lesser extent by radiation. Also, conventional heating is comparatively slow, depending on sample volume as well as the thermal conductivity of the material. MW heating is based on electromagnetic energy being converted into thermal energy. Since the long wavelength MWs can penetrate deep into the material, heat can be more uniformly generated across the sample bulk. When compared to conventional heating processes, MW irradiation-based heating can reduce power consumption by up to 97%.²³ Other favorable characteristics of the approach are noncontact heating, rapid heating, material-selective heating, volumetric heating, and a high level of safety and automation. The significant advantages of MW heating make it a powerful tool in various technological and scientific fields.

2.2. Mechanism of Microwave Heating of Carbon. The capacity of a material to be heated in the presence of MWs is defined by its dielectric loss tangent: $\tan \delta = \epsilon''/\epsilon'$, where ϵ' and ϵ'' are the dielectric constant and dielectric loss factor, respectively. The complex permittivity (ϵ) can be written in terms of dielectric constant and dielectric loss factor as $\epsilon = \epsilon' - i\epsilon''$. The dielectric constant determines how much of the incident energy is reflected and how much is absorbed, while ϵ'' is related to the dissipation of electric energy in the form of heat within the material.²⁴ MW heating takes place by the dual mechanism of ionic conduction and dipole rotation. Ionic conduction denotes the electrophoretic migration of ions under the influence of the changing electric field of the incident MWs, which generate heat due to the friction between ions and medium. Dipole rotation occurs when the molecules have permanent dipole moments aligning themselves in the same direction as the oscillating electric field.²⁵

In general terms, carbon materials are good MW absorbers, being easily synthesized (or modified) by MW heating. However, the materials show different interactions with MWs, according to their electrical conductivity:

- Conducting materials reflect the incident MWs on their surfaces and do not heat themselves. As an example, graphite and highly graphitized carbons reflect a fraction of the incident MW radiation which increases with their conductivity. In the case of carbons, their delocalized π -electrons are free to move inside material. The kinetic energy of some of their electrons increases due to the MW input, and they “jump” out of the material, resulting in the ionization of the surrounding atmosphere, and sparks or electric arcs are formed. Highly conducting materials are generally used in waveguides for MWs.
- Insulating materials are transparent to MWs. These materials have low loss factors, preventing dielectric heating. Insulators are often used in MW ovens to support the material to be heated.
- Materials with high dielectric loss absorb the incident MW energy and are easily heated. Most carbon materials are excellent MW absorbers.

3. SURFACE/STRUCTURAL DEFECTS INDUCED IN CARBON MATERIALS BY MICROWAVES

MW irradiation is a unique tool for the thermal processing of carbon materials, because it can raise temperatures up to 1000 °C within a few minutes (or even seconds, depending on the specific conditions).²⁶ It is an effective way to produce graphitic carbon from low-crystalline carbon for energy electrodes,^{27,28} and it has also been used to transform different kinds of materials (wood, cotton, paper, etc.) containing cellulose, hemicellulose, and lignin into porous nanostructured carbon materials.²⁹ The intense and controlled heating afforded by the MW approach is particularly well suited for inducing defects in graphene materials and for exploiting their potential for altering the pore structure, the surface area of the materials, and ultimately for building complex microstructures tailored for specific applications.^{30,31}

3.1. Role of Microwave-Induced Defects in Carbon Materials. MW induced defects in carbon materials are believed to play an important role in energy storage applications. Both the surface and internal structure (pores, edges, vacancies, heteroatoms, etc.) of carbon materials can be modified by controlled MW energy input. Carbon-based electrodes with engineered structural defects, hierarchical architecture, porous morphology, and high surface area usually display improved energy storage characteristics. The micro/mesoporous structure formed from defects helps to enhance ionic transport in electrochemical energy storage devices.³² The micro/mesoporous structure in carbon contributes to a high specific surface area, forming electrolyte reservoirs, which are favorable for transportation of ions. When defect-rich microporous carbon nanosheets are used in SC electrodes, they induce high capacity due to the large number of ion storage sites provided by edge defects and to pseudoadsorption ion-storage mechanisms.^{33,34} MW-generated 3D carbon nanostructures containing graphene/CNTs have also shown high performance in energy storage systems. MW treatment of CNTs can generate defects and vacancies, improving the electrochemical performance of electrodes for asymmetric SCs.³⁵

Open-ended and defect-rich CNTs have been prepared by MW hydrogen plasma processing for field emission applications. The authors considered the defective sites on CNT surfaces to serve as new active emission sites, enhancing the field emission performance.³⁶ On the other hand, some authors have suggested that MW irradiation can be used for annealing treatments which, depending on the conditions, can have the opposite effect, healing existing defects in carbon materials.³⁷ It has also been reported that MW annealing of CNTs decreases the concentration of structural defects, and by increasing the order in the carbon lattice, the thermal stability and the mechanical and electrical properties of the material are improved.³⁸ An increase in the aligned coordination of CNT structures subjected to MW treatments has also been reported. The authors suggested the alignment was reached through dipole movements in the MW-induced electric field.³⁹

Our group has also taken advantage of the ability of MW techniques to induce defects in graphene materials in a controlled manner. As an example, we have synthesized metal oxides wrapped by void-containing defective graphene using a domestic MW oven. When applied in SC electrodes, the MW-synthesized hybrids provided high cyclic stability in charge/discharge operations.⁴⁰ In another research study, nanohole-structured rGO materials with embedded Pd nanoparticles have

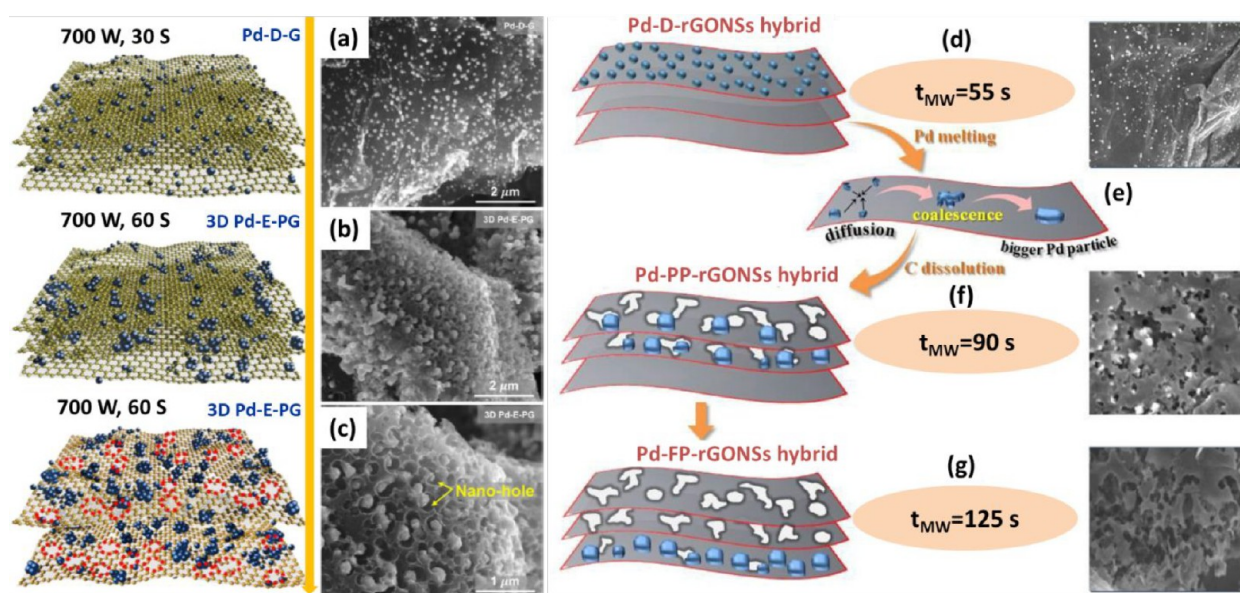


Figure 2. MW synthesis route for nanohole-structured and Pd-embedded porous rGO nanosheets. (a–c) Schematic illustration of the MW fabrication process of the 3D Pd embedded porous graphene nanosheets and the respective SEM images.⁴¹ Reprinted with permission from ref 41. Copyright 2015 American Chemical Society. (d–g) Schematic drawing showing the mechanism responsible for the formation of defective perforated Pd-rGO nanosheets along with the corresponding SEM images.⁴² Reprinted with permission from ref 42. Copyright 2017 Elsevier.

been processed by MW irradiation, yielding high-performance hydrogen storage and CO oxidation.⁴¹ Upon multistep MW reactions, agglomerated Pd nanoparticles were embedded in the basal plane of rGO sheets, as can be seen in Figure 2a–c, while for shorter heating times, much smaller palladium nanoparticles were dispersed among the layers, bonding to the rGO surfaces. In a subsequent study in the same rGO/Pd system, a controlled density of defects could be achieved by dry MW-assisted processing.⁴² During the MW treatment of rGO with Pd, high enough temperatures were achieved to allow melting of the metal and its coalescence into bigger volumes, which were then able to dissolve more carbon. Figure 2d–g shows a schematic illustration of the mechanism for defect generation in rGO as well as the corresponding SEM images of the composite.

The irradiated MW energy is continually absorbed by graphite oxide, increasing its temperature. Initially, the graphite oxide is converted into rGO nanosheets and Pd nanoparticles uniformly decorate the sheets. As the MW irradiation progresses, the Pd nanoparticles quickly reach high temperatures and are melted. In the liquid state, the Pd droplets display high mobility on the rGO nanosheet surfaces, and the small droplets coalesce into larger sized drops. When the molten Pd metal reaches defect sites in the rGO surface, it can react with the carbon, which is gradually removed from the edges of the defective region and dissolved in the liquid metal. In the liquid state, Pd is able to incorporate a very large amount of carbon. Eventually, the Pd creates micron-size holes having the dimensions of the metal droplets, and some of the perforations exhibit the shape of its path during the random-walk migration across the surface.

3.2. Simultaneous Reduction and Exfoliation of Carbon Materials. The simultaneous MW reduction and exfoliation of graphene derivatives exhibits clear advantages, when compared to traditional methods, such as the possibility of mass production of graphene-based materials with a few layers, high conductivity, and large surface area. During the reduction process, the oxygen-containing groups present on the surfaces and between the graphitic layers are removed by the interaction

with MW irradiation. According to the experimental observations, MW treatments also promote the easy attachment of metal oxide nanoparticles and conductive polymers.

Rapid heating of graphene derivatives (graphite oxide, GO) via MW irradiation generates few-layered defective graphene or rGO. The exfoliation of graphitic layers during MW irradiation reduction is a consequence of the sudden removal of oxygen-containing groups in the oxidized graphite/GO. As the MW irradiation proceeds, the threshold energies for breaking the chemical bonds between the graphitic framework and each type of oxygen-containing group is sequentially reached. The continuous transfer of energy also heats the gases (CO, CO₂) formed in the spaces between oxygenated graphitic sheets, further increasing the pressure.⁴³ The pressure exerted on the graphitic layers during exfoliation is very susceptible to the increase in temperature. At 300 °C, the pressure is around 40 MPa, reaching 130 MPa at 1000 °C. It has been estimated that nearly 2.5 MPa should be enough to separate stacked graphitic layers from graphite oxide.⁴⁴

During the MW-assisted reduction process, removal of oxygen-containing groups also removes some carbon atoms from the graphitic plane, resulting in the generation of a more distorted, defective carbon lattice, along with the release of CO/CO₂ gases.⁴⁵ Defective graphene can be prepared by MW reduction because rGO can be heated above 1000 °C upon MW irradiation.⁴⁶ Defective carbon structures formed by MW exfoliation exhibit increased surface area and porosity, which in turn helps the access of electrolytes, promoting easier ionic migration during the operation of electrodes in electrochemical devices. The defects and distortions affect the electronic properties of graphene materials by introducing scattering centers.

3.3. Heteroatom Doping. The presence of heteroatoms replacing (or bonded to) the carbon atoms in the framework of graphene is another approach to create defects in carbon materials.⁴⁷ MW irradiation generates heat energy, triggering the desired chemical reactions for doping carbon materials with

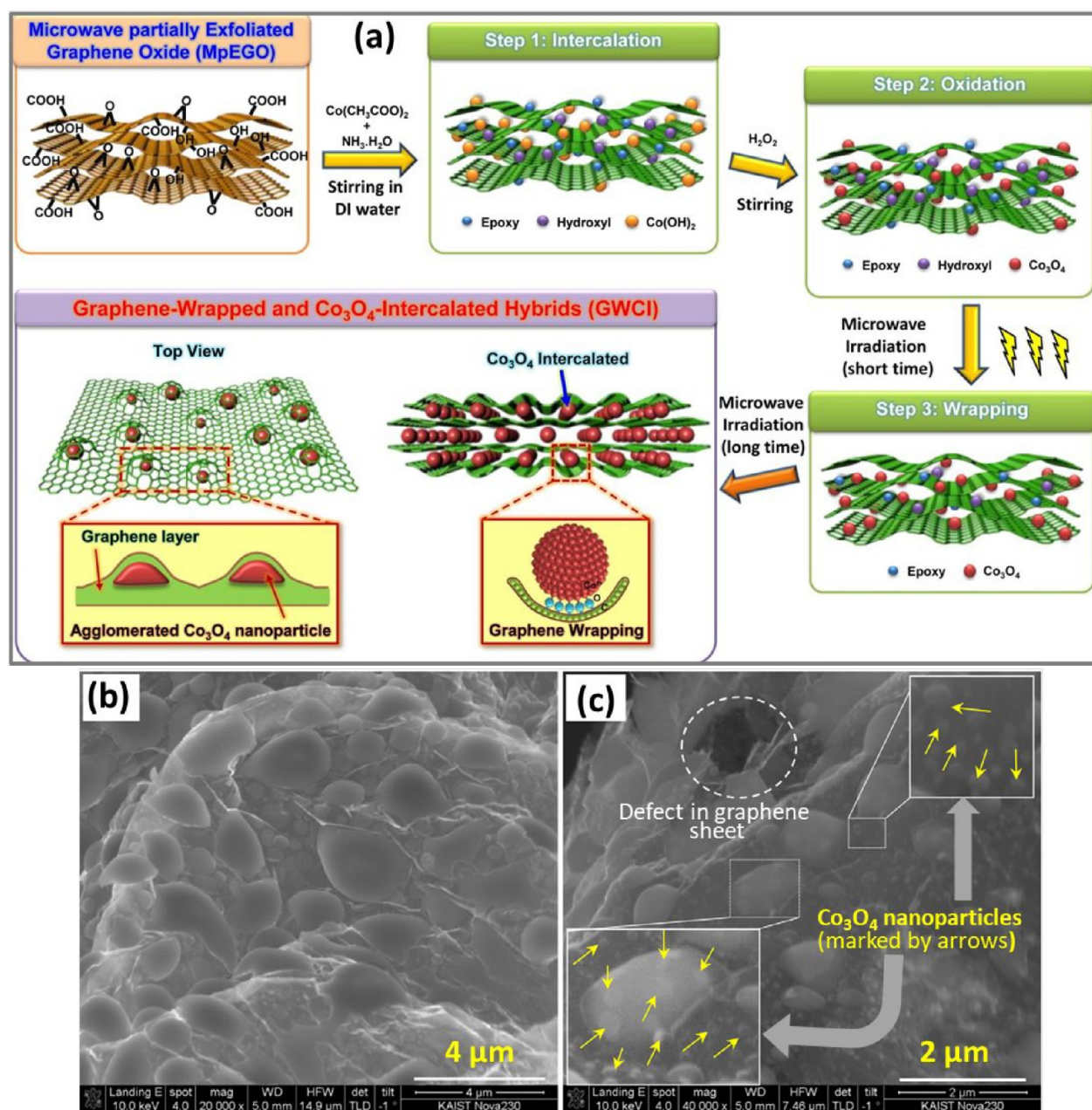


Figure 3. (a) Schematic representation of the preparation sequence for GWCI hybrids, and (b, c) SEM images of the graphene-wrapped and Co_3O_4 -intercalated hybrids.⁶⁴ Reprinted with permission from ref 64. Copyright 2014 Elsevier.

heteroatoms. Our group has recently published a review article where we detailed the effect of heteroatom doping/codoping on the properties of graphene-based materials. The review article provides an overview of the various doping strategies already reported in the scientific literature for tailoring the structure/properties of graphene-based materials.⁴⁸ Heteroatom dopants like B, N, P, and S are frequently employed in carbon structures. Among the dopants, N is the most convenient due to its similarity in atomic radius with carbon and the availability of a higher number of valence electrons (5), which are favorable for the formation of covalent bonds.⁴⁹

Taking advantage of MW irradiation, N-doped carbon quantum dots have been synthesized using citric acid and urea as carbon and nitrogen precursors, respectively. The homogeneously mixed suspension was irradiated in a commercial MW oven at 700 W for 10 min.⁴⁷ N-doped ordered mesoporous

carbon has been synthesized by a MW-assisted method using $\text{C}_3\text{H}_6\text{N}_6$ and cross-linked phenolic resins. Nickel foam coated with the precursors was heated using MWs at 1000 W/2000 W in N_2 atmosphere for 3 min to get N-doped mesoporous carbon.⁵⁰ N-doped graphene foams have been synthesized using $\text{C}_2\text{H}_5\text{OH}$, dimethylformamide (DMF), graphite, and green algae, *Chlorella vulgaris* (the N-precursor). The materials were loaded into a MW reactor and heated under pressure to 180 °C using different powers and times. After the MW treatment, the products were washed with distilled water and dried at 120 °C. Finally, the samples were carbonized at 900 °C and then heated for 4 h under a nitrogen flow.⁵¹

N-doped CNTs have been proved to offer enhanced ion storage due to the defects created by doping. The open structure allows easy diffusion of ions into the interstices between the curved carbon layers, and the N-doped sites improve the

reversible capacity of the material for charge storage.^{52,53} In our research group, N-doped CNTs have been synthesized using different carbon sources (sunflower oil and jatropha-derived biodiesel) with $\text{NH}_3/\text{CH}_3\text{CN}$ as N-containing precursors.^{54,55} These studies have shown that N-doping results in CNTs having a bambolike inner core structure, with the graphitic sheets showing degraded crystallinity. In another study, 3D hierarchical structures have been grown, containing N-doped CNTs on the surface of graphene. A bamboo-shaped internal structure was formed in the CNTs due to the presence of N atoms from the decomposition of pyridine.⁵⁶ The material exhibited high electrochemical performance, with good long-term cyclic stability in energy storage applications.

Co-doped (P and N doped) mesoporous carbon materials have been synthesized by MW-assisted carbonization of tannin cross-linked to $\text{C}_3\text{H}_6\text{N}_6$ in the presence of polyphosphoric acid for SC applications. The N content in mesoporous carbon was controlled by varying the amount of tannin in the carbonization process.⁵⁷ B–N codoped rGO has been synthesized by MW irradiation using graphite oxide, ammonia solution, and boric acid. First, graphite oxide, ammonia solution, and boric acid solution were stirred under mild heating (60 °C, 8 h), then washed using hot deionized (DI) water, and dried at 80 °C. Finally, the resulting powder was treated by MW irradiation (700 W, 40 s) to get B–N codoped rGO.⁵⁸

In our research group, we have studied N-doped-rGO synthesized using graphite oxide and $\text{C}_3\text{H}_5\text{N}$. The components were homogeneously mixed and dried, and the resulting powder was irradiated in a domestic MW oven (900 W for 1 min) for reduction and exfoliation. In this process, graphite oxide was converted into N-doped rGO, with the N being derived from $\text{C}_3\text{H}_5\text{N}$ decomposition. Finally, this N-rGO was decorated and intercalated with Pd nanoparticles by MW irradiation and studied for its electrocatalytic activity in direct-ethanol fuel cells.⁵⁹ We have also codoped rGO with N–S for application in electromagnetic interference shielding. For this purpose, graphite oxide and $\text{CH}_4\text{N}_2\text{S}$ were chosen as the carbon and N–S sources, respectively. During the MW irradiation of the mixed powder (900 W for 30 s), graphite oxide was converted into rGO, while $\text{CH}_4\text{N}_2\text{S}$ decomposed, providing N and S atoms for in situ N–S codoping of rGO.⁶⁰ The N–S codoped rGO exhibited a surface morphology almost identical to the one for pristine rGO nanosheets.

4. MICROWAVE-ASSISTED CARBON NANOSTRUCTURE-BASED COMPOSITES

Synthesis and modification of carbon materials (and composites) through MW irradiation has attained excellent results, exhibiting several advantages when compared to other fabrication methods.⁶¹ MW-assisted synthesis routes are convenient and low energy consumption methods, suitable for efficient production of carbon based composites within a greatly shortened processing time. Different types of morphologies can be achieved using a wide range of possible synthesis strategies for carbon composites (intercalated, decorated, mixed, wrapped, perforated, etc.). Many of the research works published by our group over the past few years were focused on the MW-assisted modification of carbon materials with various metal oxides to form carbon-based composites.

4.1. Nanoparticle-Supported Carbon Materials. Hybrid nanostructures composed of various metal oxide nanoparticles and nanostructured carbon materials have been prepared by MW techniques applied to dry mixtures in ambient air. As an

example, Fe_3O_4 nanoparticles have been attached to the surfaces of pristine CNTs and N-doped CNTs by MW irradiation.⁶² Optimization of the process showed that 1.5 min of irradiation with 800 W MW power was sufficient for uniformly covering the CNTs/N-doped CNTs with Fe_3O_4 nanoparticles. Iron chloride was adopted as the precursor for Fe_3O_4 nanoparticle decoration. The sizes of anchored Fe_3O_4 nanoparticles on the surface of CNTs and N-doped CNTs were 5–15 and 5–10 nm, respectively. Our subsequent research work on Fe_3O_4 decoration has shown that faceted Fe_3O_4 nanocrystals (NCs) could also be embedded into rGO nanosheets by MW irradiation.⁶³ When the mixture of precursors was irradiated at 700 W MW power for different irradiation times, distinct surface morphologies were obtained. After irradiation for 1.25 min, Fe_3O_4 nanoparticles were attached to the rGO sheets, while MW treatment for 1.75 min yielded Fe_3O_4 NCs embedded into the rGO materials.

Using a different method from the ones previously described, we have synthesized graphene-wrapped and Co_3O_4 -intercalated (GWCI) hybrid materials with just 6 min of MW irradiation.⁶⁴ MW-exfoliated GO and $\text{Co}(\text{CH}_3\text{COO})_2$ were mixed in DI water with the help of magnetic stirring and sonication. Small amounts of diluted ammonia and H_2O_2 were then added slowly to the solution. After drying in a vacuum oven at 50 °C for 1 h, the material was finally treated with 700 W MW irradiation to achieve graphene- Co_3O_4 hybrids. Figure 3a shows the complete schematic process for synthesis of GWCI hybrids, and in Figure 3b,c one can see the SEM images of the materials (Co_3O_4 nanoparticles marked by arrows in Figure 3c) which were synthesized by the process just described. Composite materials combining Co_3O_4 and rGO have also been synthesized by MW irradiation of a mixture of graphite oxide and $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$. The average size of agglomerated Co_3O_4 nanoparticles dispersed on the rGO nanosheets after being treated with a MW power of 810 W for 3.5 min was between 100 and 400 nm. The rGO nanosheets acted as anchoring sites for attachment of the agglomerated Co_3O_4 nanoparticles.

In another research work, composites including rGO nanosheets and two different types of metal oxide nanoparticles (NiO/MnO_2) have been synthesized using MW irradiation.⁶⁵ The precursors, $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and $\text{MnSO}_4 \cdot \text{H}_2\text{O}$ were mixed in a graphite oxide suspension in DI water. After drying, the resulting powder was treated by MW irradiation (800 W, 130 s). The high temperature generated from MW exposure quickly helped to decompose the $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and $\text{MnSO}_4 \cdot \text{H}_2\text{O}$ precursors into NiO and MnO_2 nanoparticles, respectively. Upon partial reduction of graphite oxide, the basic graphene framework still keeps residual oxygen functional groups, helping the attachment of uniformly distributed NiO/MnO_2 nanoparticles to rGO. Recently, we have studied the formation of honeycomblike open-edged rGO nanosheets loaded with transition metal oxides ($\text{NiO}/\text{Co}_3\text{O}_4$ nanoparticles) by MW irradiation.⁶⁶ In this instance, nickel and cobalt acetates were the precursors adopted, together with graphite oxide. The optimized MW parameters (700 W, 45 s irradiation) were able to generate a porous morphology containing $\text{NiO}/\text{Co}_3\text{O}_4$ nanoparticles more densely concentrated at the open edges of rGO nanosheets. The composite exhibited a high surface area ($\sim 570 \text{ m}^2/\text{g}$) due to the exfoliated, open edged morphology. The relatively large ($\sim 100 \mu\text{m}$) edges hold the $\text{NiO}/\text{Co}_3\text{O}_4$ nanoparticles on closely separated few-layer graphene sheets.

Ternary hybrid materials consisting of MW-exfoliated rGO nanosheets wrapping MnCo_2O_4 nanoparticles have been

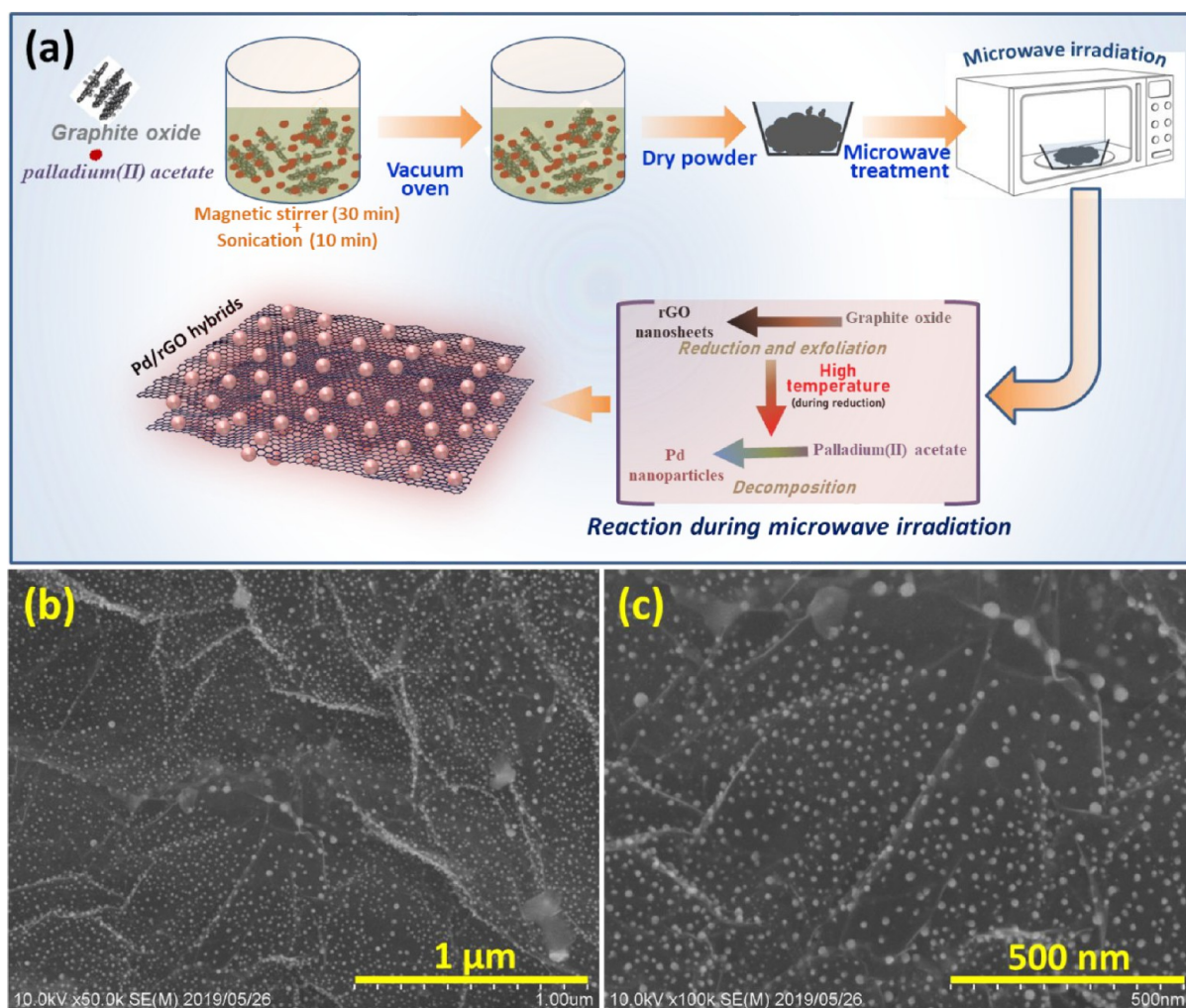


Figure 4. (a) Schematic diagram showing the steps for MW synthesis of Pd/rGO hybrids and (b, c) SEM images of Pd/rGO hybrids showing the uniform distribution of Pd nanoparticles on the rGO nanosheets.⁷¹ Reprinted with permission from ref 71. Copyright 2021 Elsevier.

synthesized by dry MW irradiation.⁶⁷ For synthesis of the composite, the initial materials were $\text{MnSO}_4 \cdot \text{H}_2\text{O}$, $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$, and graphite oxide. The metal salts were dissolved in DI water with suspended graphite oxide and then sonicated and stirred. Finally, the dried powder was treated by MW irradiation at 900 W for different times (45 s, 55 s, and 70 s). In the resulting material, the MnCo_2O_4 nanoparticles were completely wrapped by wrinkled rGO nanosheets, which provided a conductive scaffold. The material exhibits high surface area, and the appropriate pore sizes effectively favor the migration of electrolyte ions inside its bulk. We have studied the production of binary metal oxides (Mn_3O_4 – Fe_2O_3 / Fe_3O_4) anchored to rGO by MW irradiation.⁶⁸ Graphite oxide, manganese(III) oxide, and iron(III) oxide were dispersed in $\text{C}_2\text{H}_5\text{OH}$ using sonication and magnetic stirring. After evaporation of the alcohol, the dried powder was irradiated with MWs at constant power (900 W) for 30 s, in order to obtain Mn_3O_4 – Fe_2O_3 / Fe_3O_4 @rGO ternary hybrids. The ternary hybrids exhibited a specific surface area of $\sim 322 \text{ m}^2/\text{g}$.

Yokoyama et al. have studied the dissolution and diffusion of graphitic carbon in palladium.⁶⁹ At high temperatures, the equilibrium can be described as (eqs 1 and 2):



Therefore, carbon dissolves into the metal up to the solubility limit, as long as graphite is present. The carbon dissolution can be exploited in MW treatments, leading to the formation of holes in the graphite/graphene layers in contact with the metal. The resulting open structure is considered to be largely beneficial for electrochemical devices, since it enhances the migration of electrolyte ions inside the material.

The Pd-rGO combination proved to be very versatile, enabling the fabrication of different morphologies, depending on the exact MW synthesis conditions. Besides yielding Pd nanoparticle-attached rGO, the method also enabled the Pd catalyst-assisted growth of CNTs on rGO composites synthesized by MW irradiation.⁷⁰ In this instance, the graphite oxide powder was mixed with palladium(II) trifluoroacetate ($\text{Pd}(\text{TFA})_2$) or palladium(II) acetate ($\text{Pd}(\text{OAc})_2$) for the dry MW synthesis. The two types of mixtures with graphite oxide (differing in the Pd salts) were separately treated by MW irradiation (810 W) for 1.5 min or for 3.0 min, leading to the formation of Pd-attached rGO and Pd catalyst-assisted growth of CNTs on rGO, respectively. In yet another work, we have uniformly decorated rGO nanosheets with Pd nanoparticles by the MW method shown in Figure 4a.⁷¹ For synthesizing the Pd/rGO hybrids, $\text{Pd}(\text{OAc})_2$ was mixed with graphite oxide in

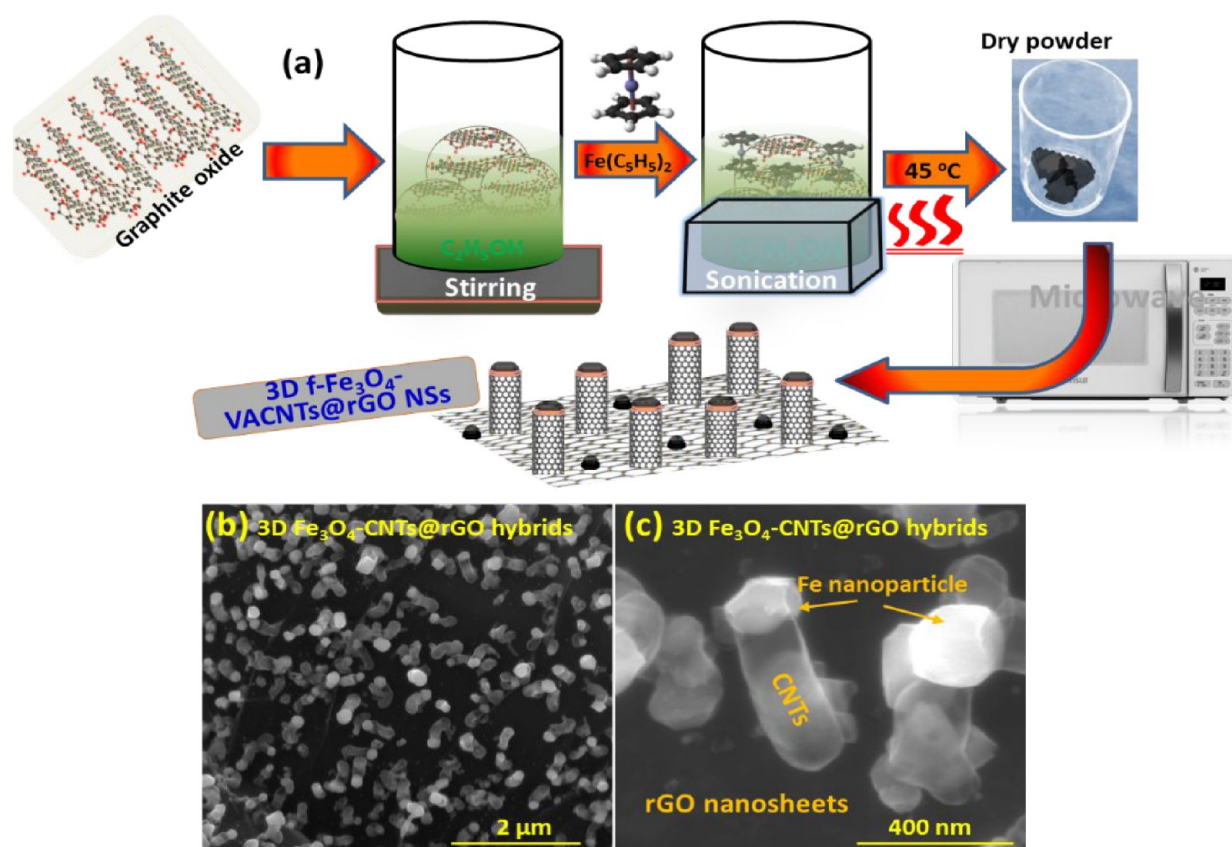


Figure 5. (a) Schematic representation of the synthesis of 3D faceted Fe_3O_4 -CNTs@rGO hybrids and (b, c) SEM images of the resulting material.⁷⁷ Reprinted with permission from ref 77. Copyright 2019 Elsevier.

$\text{C}_2\text{H}_5\text{OH}$ and dried at 40 °C for 10 h to evaporate the alcohol. The dried powder was treated by MW irradiation using a power of 700 W for 20 s for decomposition of $\text{Pd}(\text{OAc})_2$ into Pd nanoparticles, reduction and exfoliation of graphite oxide into rGO nanosheets, and formation of Pd/rGO hybrids. In the SEM images in Figure 4b,c, one can easily notice the uniform distribution of Pd nanoparticles (anchored to the basal plane of rGO) achieved in the composite. The slightly higher concentration of particles attached to the edges of the rGO sheets are logically expected, due to the inherently larger number of reactive sites.

Deposition of thin layers of ZnO on graphene sheets via MW irradiation (700 W, 5 min) has been obtained using $\text{Zn}(\text{NO}_3)_2$ and graphite oxide.⁷² During the MW treatment of the precursor mixtures, graphite oxide is converted into porous GO, and $\text{Zn}(\text{NO}_3)_2$ is simultaneously decomposed into Zn^{2+} ions. As the treatment progresses, the GO starts to be reduced, and the negative functional groups on the porous GO react with the Zn^{2+} ions, leading to the formation of Zn–O–C and Zn–O bonds which attach the Zn atoms to the rGO sheets. The ZnO nanoparticles were uniformly deposited on the rGO surfaces. In a recent study, we have demonstrated the synthesis of ZnO nanoparticles attached to transparent rGO nanosheets via direct MW irradiation.⁷³ In this case, $\text{Zn}(\text{CH}_3\text{CO}_2)_2 \cdot 2\text{H}_2\text{O}$ was chosen as the source of Zn, and in the MW irradiation process, the powder mixture was treated at a lower power (500 W) and shorter irradiation time (90 s). The ZnO nanoparticles were distributed over the rGO nanosheets without aggregation. The changes in the precursor for ZnO nanoparticles and its concentration played a crucial role during the deposition of

ZnO nanoparticles on rGO. Also, the variations in the MW power and irradiation time provided the means to control the sizes of the ZnO nanoparticles as well as the final configuration of the hybrid material. In this way, one could obtain different composite morphologies, varying from dispersed nanoparticles decorating the rGO until full ZnO film coverage.

4.2. 3D Carbon Nanostructures with Metal Oxides.

Recently, 3D carbon nanostructures containing graphene-based materials and CNTs have attracted much attention, since they provide high surface area and electrical conductivity with open porosity. The 3D carbon nanostructures show significant mechanical strength along out-of-plane directions, which is something not easy to achieve with 2D stacked graphene sheets or 1D CNTs by themselves. We have used MW irradiation (800 W, 4 min) to fabricate self-assembled 3D hierarchical carbon structures containing Co_3O_4 nanobeads, CNTs, and GNSs.⁷⁴ The starting materials for the large, gram scale synthesis, were $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and graphite oxide. The simple one-step method simultaneously induced the formation of Co_3O_4 nanobeads and CNTs on graphene nanosheets, providing a uniform dispersion of the CNTs and Co_3O_4 . Also in our group, rGO nanosheets associated with different metal oxides (CoO , NiO and NiCoO_2) have been synthesized for a comparative study of field emission performance.⁷⁵ The mixtures of GO powder with metal salts ($\text{Co}(\text{OAc})_2 \cdot 4\text{H}_2\text{O}$ and $\text{Ni}(\text{OAc})_2 \cdot 2\text{H}_2\text{O}$) were treated by MW irradiation. The MW treatments performed at fixed power (810 W) for 3 min enabled the formation of rGO- CoO , rGO- NiO , and 3D rGO- NiCoO_2 composites.

We have synthesized interconnected 3D networks of Fe_3O_4 /rGO nanosheets in which the open edges of rGO were filled with Fe_3O_4 nanoparticles.⁷⁶ In the synthesis, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ and graphite oxide were mixed by stirring and sonication in $\text{C}_2\text{H}_5\text{OH}$. After ethanol evaporation in an open atmosphere, the dried powder was MW treated for obtaining 3D Fe_3O_4 /rGO hybrids. We found that the Fe_3O_4 particles (with well-defined faceted shapes) nucleated in the open spaces among rGO nanosheets, suppressed their restacking, thus inducing a configuration where the Fe_3O_4 nanoparticles remain separated by rGO nanosheets and are free from agglomeration. This kind of graphene-based 3D hybrid material with open pores decorated by pseudocapacitive particles has displayed superior performance for application in SC electrodes. After further research, this study led to the development of self-assembled 3D hybrid nanostructures with uniform CNTs, having Fe_3O_4 nanoparticles on their tips, growing perpendicular to the surfaces of rGO nanosheets.⁷⁷ Once again, the material has been obtained by a dry MW-assisted approach, using $\text{Fe}(\text{C}_5\text{H}_5)_2$, graphite oxide, and $\text{C}_2\text{H}_5\text{OH}$ as starting materials. The mixture was stirred, sonicated, and after drying was directly treated by MW irradiation (900 W for 2 min) for the synthesis of 3D Fe_3O_4 -CNTs@rGO hybrids, as shown in Figure 5a. The vertical CNTs grow with the help of the Fe catalyst, and after oxidizing the Fe nanoparticles, the growth was terminated, resulting in the formation of small size (<500 nm) CNTs with Fe_3O_4 remaining on their tips, as shown in the SEM images in Figure 5b,c. The carbon species from rGO diffuses into the Fe catalyst, contributing to the growth of CNTs on the rGO surfaces.⁷⁸

We have grown 3D Ga_2O_3 nanowormlike structures on GNSs by the MW-assisted method using a solid mixture of graphite oxide and gallium acetylacetonate.⁷⁹ The Ga_2O_3 wavy rods grew on the rGO nanosheets with an average diameter of 200 nm and lengths of up to 1 μm . MW irradiation time and power were found to greatly influence the morphology of the hybrid materials. The reaction is fast, involving only 1–3 min of MW irradiation under atmospheric pressure. MW irradiation growth of ZnO-GO heterostructures has also been achieved by our group using graphite oxide and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ as the initial materials.⁸⁰ By varying the MW irradiation times at a fixed irradiation power of 800 W, we have obtained heterostructures supported on GO nanosheets composed of 3D ZnO microcubes, 2D nanoflakes, or 1D ZnO nanoneedles after irradiation for 3.0, 4.5, and 7.5 min, respectively.

5. MICROWAVE-ASSISTED FABRICATION OF CARBON-BASED MATERIALS FOR ENERGY STORAGE ELECTRODES

The MW synthetic approach, with its great potential (discussed in the previous sections), has had a huge impact on energy storage research. With rapid progress in this field, carbon materials have been extensively combined with different metal/metal oxides/mixed metal oxides, metal sulfides/mixed metal sulfides, metal hydroxides/layered double hydroxides, other carbonaceous materials, etc. using various MW routes. Such kinds of MW-synthesized carbon materials have been extensively investigated for various applications, including EMI shielding, fuel cells, field emission devices, catalysts, sensors, hydrogen storage, etc.^{81–84} Among the electrochemical energy storage systems, SCs and batteries are considered to be the top devices, in terms of performance and development potential. As expected in this context, the MW approach has

been extensively applied for development of both SC and battery electrodes.

5.1. Supercapacitor Electrodes. Although SCs are not as widely recognized as the ubiquitous batteries, they are probably the most promising energy storage devices of the current century, owing to their enhanced energy storage ability, long cycle life, and superior power density. Carbon materials usually store charges by an electric double layer capacitor (EDLC) type mechanism. High surface area, enhanced conductivity, and porous architecture are the basic requirements to become a suitable SC electrode. In this aspect, it is important to mention that the MW-assisted exfoliation and reduction of carbon materials is quite able to enhance these properties. Moreover, carbon materials serve as good receptors for MW absorption, and the MW processing of GO is one of the key strategies for producing SC electrodes.⁸⁵ As an example, the rapid MW treatment of GO along with graphene nanoplatelets was found to be favorable for the development of flexible solid state SC devices.⁸⁶ Recently, a MW-assisted reduction process has been adopted to produce graphene frameworks with high porosity (a composite of holey graphene and carbon dots), which achieved a high surface area (1950 m^2/g) and enhanced conductivity (6230 S/m).⁸⁷ As a result, the porous graphene material displayed exceptional electrochemical performance.

The MW reduction of GO/graphite oxide has been carried out either with the raw materials in a dry state or dissolved/suspended in a liquid. For treatments performed on carbon materials in liquid media, ionic liquids have generally been preferred. As an example, the ionic liquid 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (EMIM-NTf₂) has been used to exfoliate and reduce graphite oxide to produce rGO.⁸⁸ The corresponding rGO-based SC device displayed superior energy density within an extended potential limit of 3.5 V. It is important to mention that the ionic liquid also served as a good absorber of MW radiation due to the presence of ions with high polarizability.

We have already discussed the creation of defects in graphene and the utilization of ionic liquids in MW treatments. Combining these two concepts, researchers were able to grow CNTs on graphene through the MW synthetic route.⁸⁹ This approach is comprised of three steps: (1) MW-assisted defect creation on graphene nanoplatelets, (2) affixing metal nanoparticles into the defects, and (3) growth of CNTs from the metal nanoparticles with the help of an ionic liquid. One of the major advantages of this approach is the strong anchoring of the CNTs to the graphene substrates, just like the “tip growth mechanism” in the chemical vapor deposition process of CNT growth. Such a strong interconnection between these two allotropes of carbon enables the 3D nanostructure to achieve enhanced performance in electrochemical devices. The 3D graphene/CNT/Pd composites (exhibiting a surface area of 397.20 m^2/g) developed by Sridhar et al. for application in SC electrodes, exhibited a remarkable specific capacitance of 1615 F/g at a scan rate of 10 mV/s (in 1 M KOH electrolyte solution), and a capacitance retention of 146% after 1000 cycles, which is a significant increase over the original value.⁹⁰ It is important to remember that the presence of vertically aligned CNTs in the 3D composite not only enhances the capacitive performance, through its EDLC contribution, but also prevents the restacking of the graphene sheets, keeping the high surface area easily accessible to the electrolyte ions.

Inspired by the effective strategy for CNT growth through the MW treatment of graphene materials with nanoparticles, our

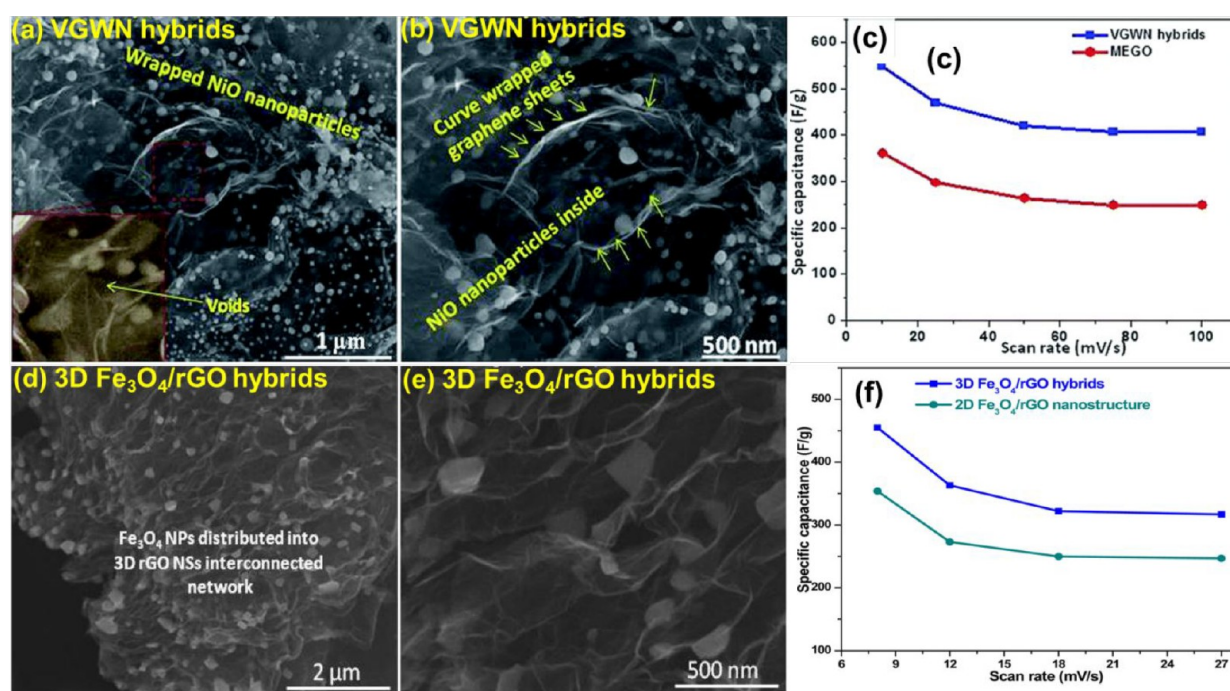


Figure 6. Morphology and electrochemical performance of composites based on metal oxides and MW exfoliated rGO: (a, b) SEM images and (c) specific capacitance at different scan rates of rGO/NiO composites.⁴⁰ Reprinted with permission from ref 40. Copyright 2016 Royal Society of Chemistry. (d, e) SEM images and (f) specific capacitance at different scan rates of rGO/Fe₃O₄ composites.⁷⁶ Reprinted with permission from ref 76. Copyright 2017 American Chemical Society.

research group has developed SC electrodes based on a 3D composite of CNTs, GNSs, and Co₃O₄.⁷⁴ However, when compared to the previously discussed research from reference, our synthesis process is considerably cheaper, since we have not used any costly ionic liquid. Moreover, the Co₃O₄ nanoparticles were not only deposited on the graphene sheets but also anchored to the CNTs, forming a “necklacelike” structure. As a result, the strong synergistic contribution from the three components enhanced the capacitive properties of the composite. Additionally, the 3D carbon nanostructures of CNT and graphene acted as a reliable conductive channel between the pseudocapacitive materials and the current collector, improving the cycle life of the electrodes (capacitance retention of 94.5% after 5000 cycles in KOH electrolyte solution). Overall, this MW approach can be considered as an efficient way to develop 3D SC electrodes from 0D metal oxides, 1D CNTs, and 2D graphene nanosheets.

Taking into account the results previously discussed, it is clear that carbon materials are excellent substrates for incorporation of metal/metal oxide nanoparticles. Our research group has successfully created voids in graphene sheets through the partial MW exfoliation of GO and further incorporated NiO nanoparticles into those voids.⁴⁰ When compared with conventional heating, the MW heating process is highly efficient for generating holes/void spaces in the carbon nanostructures. Such voids are favorable for the easy diffusion of electrolyte ions. Moreover, the composite morphology with the graphene sheets wrapping the nanoparticles prevents the structural deformation of the material during continuous charging/discharging processes, leading to a long cycle life.

The reaction time plays a very important role in the final outcome of MW heat treatments. The morphology of the composite is progressively changed during the treatment, and we have already mentioned the excellent electrochemical character-

istics of electrodes treated for short times. However, when treatments are performed over longer periods, the continuous irradiation of the material with high-energy MWs gradually leads to the agglomeration of nanoparticles, because surface diffusion takes place at an accelerated rate, given the high temperatures generated from the MW input. In this context, the partial exfoliation of GO through MW treatment was found to be favorable for the intercalation of metal oxide nanoparticles into graphene sheets. The wrinkled, curly 3D network of graphene sheets in partially exfoliated GO not only enhances the surface area of the material but also works as a suitable environment for wrapping other nanomaterials. Moreover, the porous nature of this graphene material was found to be favorable for easy transport of electrolyte ions. When compared to conventional hydrothermal/solvothermal approaches, the MW method has intrinsic potential for faster, more effective, and controlled creation of defects in graphene sheets.

Our research group has adopted a MW methodology to synthesize SC electrodes with high cycling stability. Through a dry MW approach, Co₃O₄ nanoparticles were successfully intercalated among the graphene sheets, retaining a surface area of 331.2 m²/g with a pore volume of 1.23 cm³/g.⁶⁴ Furthermore, the exterior wrapping of graphene sheets also enhanced the structural integrity of the composite, inhibiting cracks and the structural breakdown of the electrode material, resulting in the long cycle life of the SCs (155% capacitance retention after 10000 cycles using 0.5 M BMIM-BF₄/CH₃CN electrolyte solution). In one of our previous articles, a similar MW-assisted synthetic approach has been adopted to deposit ZnO nanoparticles on graphene sheets.⁷² When compared to conventional methods, the MW treatment has the advantage of depositing metal oxide nanoparticles in both the exterior as well as the less easily accessible graphene sheets in the interior of the material. The ZnO/MW-exfoliated graphene hybrid exhibited a specific

capacitance of 472 F/g at a current density of 3A/g in 1 M Na₂SO₄ electrolyte solution. Apart from this, the composite also displayed superior optical properties.

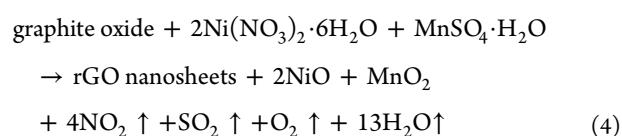
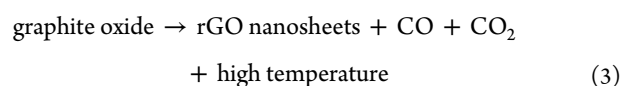
The types of precursors chosen for the generation of metal oxide nanoparticles also play an important role on the final morphology and properties of the carbon matrix composites. In the previously mentioned study, Zn(NO₃)₂ was used as the metal oxide precursor, with good results. In a more recent research, we have synthesized ZnO/rGO composites through a similar MW approach, but this time we used Zn(CH₃COO)₂·2H₂O as the precursor.⁷³ When compared to the previously reported ZnO-based composites, the composite synthesized from the acetate displayed a lower specific capacitance (102.4 F/g) using 0.1 M KOH electrolyte solution. A uniform distribution of ZnO nanoparticles has been clearly observed from SEM and HR-TEM analysis; however, in the latter composite, the ZnO layer comprised both small and large-sized particles, which were detrimental to the charge storage performance.

Apart from the metal oxides already mentioned, the MW approach has been employed to incorporate a wide variety of metal and metal oxide nanoparticles for graphene-based composites. In one of our previous articles, we have demonstrated the fabrication of 3D porous rGO, which served as an effective substrate to incorporate Fe₃O₄ nanoparticles.⁷⁶ The 3D Fe₃O₄/rGO composite exhibited good electrochemical performance within a wide potential window of 1.4 V. Most importantly, during charge/discharge cycling, the composite displayed an initial capacity fading of 8.6% (over 190 cycles) in 2.0 M KOH electrolyte solution, but after this initial drop, the capacity was maintained for 9500 cycles, indicating very good cycling stability. Such long-term cyclic performance has been attributed to the stable 3D interconnected network structure of the composite, which provides easy pathways for electrolyte diffusion.

Figure 6 shows the morphology and electrochemical performance (represented by the change in specific capacitance with respect to the scan rate) of composites of metal oxides (NiO and Fe₃O₄) and MW exfoliated rGO.^{40,76} The corresponding SEM images clearly demonstrate the creation of voids on the graphene sheets as well as the incorporation of metal oxide nanoparticles into those voids. It is important to remember that the clustering of metal oxide nanoparticles is one of the major problems for hydrothermal/solvothermal synthetic methods, and this aggregation can be avoided through well designed MW processing. The SEM images in Figure 6 show metal oxide grains uniformly distributed in rGO matrix composites, providing further support to this observation.

Recently, an *in situ* MW irradiation technique has been adopted to synthesize a mesoporous composite of rGO and α-MoO₃ with a specific surface of 21.65 m²/g.⁹¹ Displaying a porous nanoarchitecture, the composite yielded a maximum capacitance of 483 F/g at a current density of 1 A/g using 1 M aqueous H₂SO₄ electrolyte. In another work, a MW-assisted chemical precipitation method has been employed to combine rGO and MnO₂.⁹² Benefitted by its structural stability, the composite displayed better cycling performance (capacitance retention of 99.4% after 5000 cycles in 1 M Na₂SO₄ aqueous electrolyte solution) and a specific capacitance of 140.3 F/g. Different MW irradiation processes have been applied in these two research works; however, both have demonstrated the ability to improve the electrochemical performance of composites based on metal oxide nanoparticles and graphene materials.

In the constant search for further improvement of the properties of energy storage devices, MW-exfoliated rGO nanosheets have been combined with more than one type of metal oxide to form ternary composites. In this regard, our research group has demonstrated the electrochemical performance of ternary hybrid electrodes based on rGO, NiO, and MnO₂.⁶⁵ In this work, a general mechanism for MW synthesis of the ternary composite has been proposed. Initially, the graphite oxide was converted to exfoliated rGO nanosheets, with the generation of high temperature and emission of gases like CO and CO₂. Following the fast increase in temperature upon continuing MW irradiation, the decomposition of precursors [Ni(NO₃)₂·6H₂O and MnSO₄·H₂O] should occur roughly but simultaneously with the thermal reduction of graphite oxide and the growth of metal oxide nanoparticles should happen immediately after. Loosely expressed in terms of chemical reactions, eqs 3 and 4 are believed to be responsible for the formation of metal oxides as well as the reduction and exfoliation of graphite oxide, providing an overall view of composite formation:



At this point, it is important to remember that a simple mixture of metal oxide precursors could not be used for the synthesis of nanoparticles through MW irradiation without the support of graphite oxide. The presence of graphite oxide is essential, since it acts as the transducer of MW energy into heat for the reactions, playing a role similar to a susceptor in induction heating furnaces. The ternary electrodes exhibited specific capacity of 165.7 mAh/g and a cycling stability of 83.2% after 2000 cycles (in 0.1 M KOH electrolyte solution). Their electrochemical performance can be credited to the mesoporous structure (specific surface area 60.64 m²/g), high conductivity, enhanced surface area, structural stability, and synergistic effects among the components.

Composites comprising a highly conductive 3D graphene/rGO network decorated with pseudocapacitive materials are extremely well suited for application in SC electrodes. In one of our recent articles, we have studied the synthesis and SC application of rGO-enclosed NiO/Co₃O₄ nanoparticles.⁹³ Due to excellent pseudocapacitive contribution from both oxides, this porous ternary composite achieved a superior specific capacitance of 910 F/g at the scan rate 20 mV/s in 0.1 M KOH electrolyte. The electrode also displayed a decent capacitance of 298 F/g at a high scan rate (100 mV/s). The composite retained 88.9% of its initial capacitance after 2000 cycles, indicating good cycling stability. In another work, rGO nanosheets have been combined with Mn₃O₄–Fe₂O₃/Fe₃O₄ nanoparticles through MW treatments.⁹⁴ Having a specific surface area of 322 m²/g, this composite exhibited the capacitance of 590.7 F/g at the scan rate of 5 mV/s in 1.0 M KOH electrolyte solution. However, when compared with our previously reported rGO/metal oxide ternary composites, this electrode showed a lower cycling stability (capacitance retention of 64.5% after 1000 cycles).

Mixed transition metal oxides have been extensively researched as components of composite SC electrodes. In this

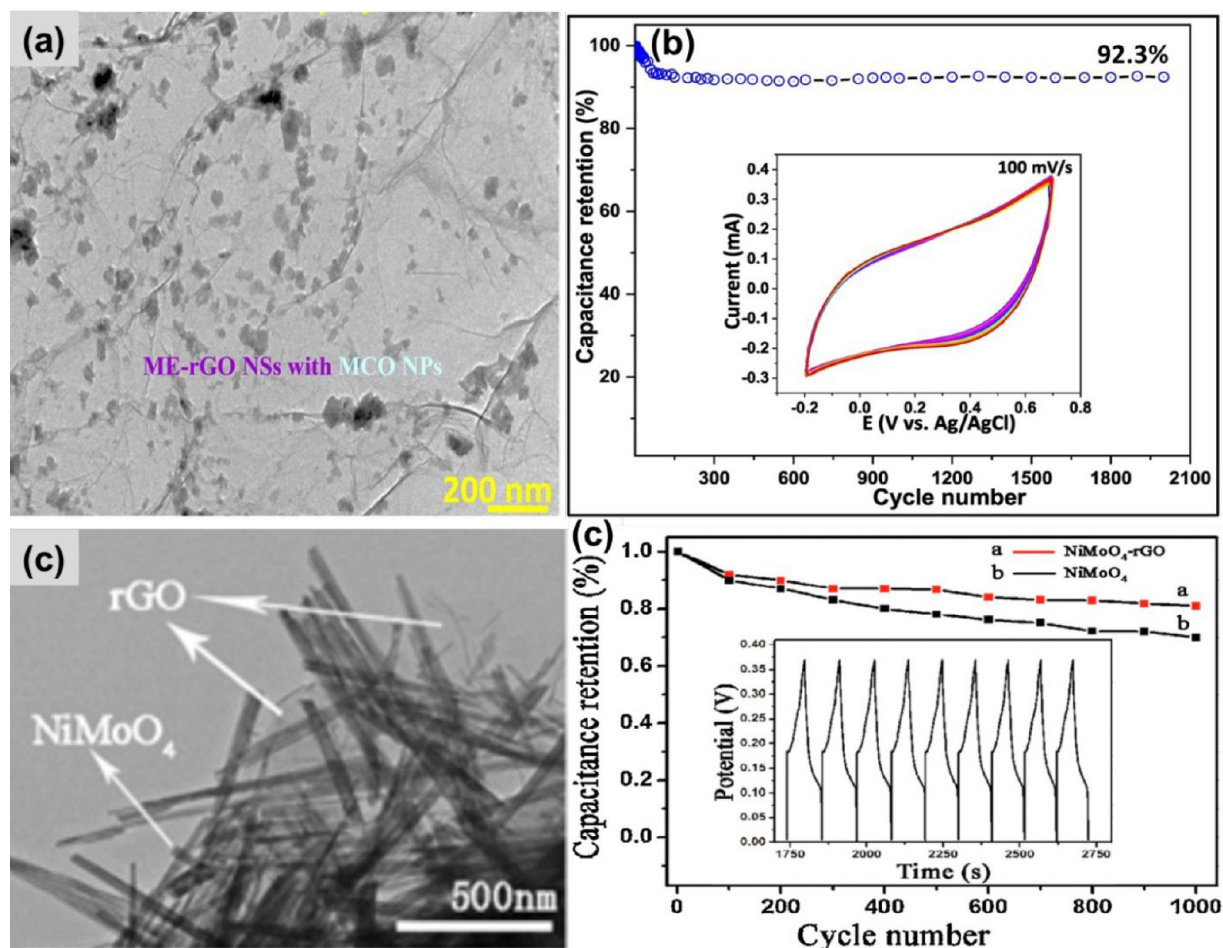


Figure 7. Morphology and electrochemical performance of MW-synthesized composites based on mixed transition metal oxides and rGO: (a) TEM image and (b) cycling stability testing for MnCo_2O_4 (MCO)/MW-exfoliated rGO (ME-rGO) composite.⁶⁷ Reprinted with permission from ref 67. Copyright 2019 Elsevier. (c) TEM and (d) cycling stability of NiMoO_4 /rGO composite.⁹⁵ Reprinted with permission from ref 95. Copyright 2015 Elsevier.

context, our group developed a MW synthesis method to fabricate SC electrodes based on MnCo_2O_4 nanoparticles wrapped with rGO nanosheets.⁶⁷ The conductive wrapping was found to be beneficial for the electrochemical performance of the material. As a result, the mesoporous composite with specific surface area of $21.66 \text{ m}^2/\text{g}$ displayed higher specific capacitance (562 F/g at the scan rate of 20 mV/s in 0.1 M KOH electrolyte solution) than our previously reported metal oxide-based binary composites. In fact, the material displayed better capacitance than MnCo_2O_4 /rGO composites prepared by other authors using solvothermal/hydrothermal routes. Liu et al. demonstrated the growth of NiMoO_4 nanorods on the surface of rGO sheets for application in high-performance SC electrodes.⁹⁵ Taking advantage of its mesoporous structure with surface area of $50.8 \text{ m}^2/\text{g}$, the electrode displayed a specific capacitance of 1274 F/g at a current density of 1 A/g in 6 M KOH electrolyte. For an easier comparison of the last two materials discussed, Figure 7 shows TEM images and the corresponding cycling stability curves for rGO-based composites with MnCo_2O_4 and NiMoO_4 .^{67,95} As one can observe in Figure 7b, both the shapes of the cyclic voltammetry curves and of the charge/discharge curves have not been significantly altered by continuous charging/discharging for a long period. The decrease in cycling stability over the initial few cycles is generally associated with the activation of electrode material. However, when comparing the

cycling stability curves in Figure 7b,d, the change in capacitance with the number of charge/discharge cycles is clearly more stable for the MnCo_2O_4 /rGO composite than the NiMoO_4 -based material. The good wrapping of metal oxide nanoparticles by the rGO sheets and the formation of an interconnected conductive rGO framework could be the reasons for the enhanced cycling stability.

Besides graphene, other carbon materials like CNTs have also been MW-treated for application in SC electrodes. For improving cycling stability, ordered mesoporous carbon has been combined with CNTs to produce core-shell structured composites.⁹⁶ The ordered mesoporous carbon not only acted as the MW absorbing component but also supplied the metal nanoparticles to catalyze the CNT growth. Because of the porous nanoarchitecture (specific surface area, $801 \text{ m}^2/\text{g}$; pore volume, $1.16 \text{ cm}^3/\text{g}$) and synergistic contributions, the composite exhibited superior electrochemical performance, including a cycling stability of 99.32% after 10000 cycles using $1 \text{ M H}_2\text{SO}_4$ aqueous electrolyte solution. In another example, CoMoO_4 nanoplatelets have been anchored to CNTs through a solvent-free MW route, where the CNTs served as the MW susceptor.⁹⁷ This mixed metal oxide-based graphene composite exhibited decent electrochemical performance, due to the combination of pseudocapacitive (CoMoO_4) and EDLC-type (CNT) materials. In yet another work, a MW-assisted

Table 1. Comparison of Electrochemical Properties for Different Graphene-Based Materials Synthesized by MW Methods for SC Application (Measured in a Three-Electrode System)

materials	synthesis method	electrolyte	specific capacitance	cycling stability	ref
rGO/MnCo ₂ O ₄	MW irradiation	0.1 M KOH	562 F/g at 20 mV/s	92.3% after 2000 cycles	67
graphene/CNT/Co ₃ O ₄	MW irradiation	30 wt % KOH	600 F/g at 0.7 A/g	94.5% after 5000 cycles	74
3D graphene/CNT/Pd	MW irradiation	1 M KOH	1615 F/g at 10 mV/s	146% after 1000 cycles	90
rGO/NiMoO ₄	solvothelmal-assisted MW	6 M KOH	1274 F/g at 1 A/g	81.1% after 1000 cycles	95
CNT/CoMoO ₄	dry MW irradiation		170 F/g at 0.1 A/g	93.2% after 1000 cycles	97
CNT/NiMn ₂ O ₄	MW-assisted hydrothermal	6 M KOH	915.6 F/g at 1 A/g	93% after 5000 cycles	98
CNT/NiO/NiS	MW irradiation		809.7 F/g at 1 A/g	100% after 20 000 cycles	99
graphene/Zn _{1-x} Ni _x MoO ₄)	MW irradiation	2 M KOH	555.5 F/g at 1 A/g	85% after 5000 cycles	100
rGO/(Co,Mn) ₃ O ₄	MW irradiation	6 M KOH	1004 F/g at 3 mV/s	96.9% after 3000 cycles	101
activated graphene foam/Co ₉ S ₈	MW-assisted hydrothermal	6 M KOH	1150 F/g at 5 mV/s	94% after 1000 cycles	102
rGO/MoS ₂	MW heating	1 M HClO ₄	265 F/g at 10 mV/s	92% after 1000 cycles	103
rGO/NiS	MW-hydrothermal	2 M KOH	1745.67 F/g at 1 A/g	54.2% after 1500 cycles	104
graphene/nickel cobalt sulfides	MW irradiation	3 M KOH	1186 F/g at 1 A/g		105
N-doped graphene/NiS	MW irradiation	6 M KOH	1467.8 F/g at 1 A/g		106
3D S-doped graphene/NiFeS ₂	MW alcohothermal process	6 M KOH	1073.2 F/g at 1 A/g	90.4% after 2000 cycles	107
graphene/Zn ₅ (OH) ₈ C ₁₂ ·H ₂ O/Ni foam	MW-assisted hydrothermal	4 M KOH	836 F/g at 1 A/g	92% after 5000 cycles	108
rGO/NiCo LDH	MW irradiation	6 M KOH	1622 F/g at 5 mV/s	85% after 5000 cycles	109
graphene oxide/Co–Al LDH	MW-assisted reflux method	6 M KOH	772 F/g at 1 A/g	73% after 10 000 cycles	110
rGO/Ni–Al LDH	MW irradiation	6 M KOH	1630 F/g at 1 A/g	80% after 2000 cycles	111
graphene/PEDOT	MW heating	1 M H ₂ SO ₄	270 F/g at 1 A/g	93% after 10 000 cycles	112
graphene/SnO ₂ /PANI	MW irradiation	1 M Na ₂ SO ₄	1012 F/g at 4 A/g	91% after 1500 cycles	113
rGO/MnN	MW irradiation	1 M Na ₂ SO ₄	639.2 F/g at 10 mV/s	95.9% after 2000 cycles	114
graphene/MoTe ₂	MW irradiation	1 M Na ₂ SO ₄	434 ± 37.5 F/g at 1 A/g	125% after 5000 cycles	115

hydrothermal process has been adopted to combine CNTs with NiMn₂O₄.⁹⁸ The corresponding composite (specific surface area up to 219.3 m²/g) not only displayed better capacitance (915.6 F/g) but also exhibited promising cycling stability of 93% after 5000 charge/discharge cycles using 6 M KOH electrolyte. Moreover, the asymmetric SC devices with activated carbon as the negative electrode showed a capacitance retention of 82.8% after 10000 cycles.

Extending the mixed oxide strategy already applied to graphite oxide and GO, the MW synthetic approach has been implemented to synthesize ternary composites based on NiO, NiS, and CNTs.⁹⁹ The ternary composite electrodes displayed no change in capacitance after 20000 charge/discharge cycles. The introduction of NiS nanoparticles was considered the main factor for enhancing the cycling stability. This electrode architecture has once again been proved favorable for energy storage applications. Table 1 shows a comparison of the electrochemical properties of different graphene-based materials synthesized by MW-assisted methods for use in SC electrode applications.

5.2. Secondary Batteries. Over the last decades, there was a clear shift in the main research directions in the field of energy, with much effort and resources been diverted to rapidly enhancing the efficiency of renewable energy devices.¹¹⁶ Secondary batteries (rechargeable batteries) are considered to be indispensable weapons in this silent revolution occurring in the global energy panorama. In order to address the ever growing demands placed on the performance of energy storage devices, it is still necessary to increase the energy density of secondary batteries.

Among the different kinds of secondary batteries, Li-ion batteries (LiBs) are found to be the most investigated. According to the working mechanism of LiBs, their performance majorly depends on the process of insertion/deinsertion of Li⁺ ions. Carbon-based materials are by far the most used

constituents in LiB anodes. As an example, advanced LiB anode materials have been fabricated by combining polydopamine-coated silicon with electrochemically exfoliated graphene in a commercial MW oven.¹¹⁷ The graphene material worked as the MW absorber, and as a result of heating, some defects were created. Due to porous structure and N-doped carbon, the composite exhibited reversible capacity (1744 mAh/g at the current density of 0.1 A/g) and good cycling stability (98.1% Coulombic efficiency over 200 cycles). The enhanced electrochemical performance has been attributed to the porous nanoarchitecture and conductive network of graphene and the coating of N-doped carbon which decreases the volume expansion of the composite during cyclic testing.

Metal oxides have been frequently combined with graphene materials to develop high-performance LiB anodes due to their enhanced theoretical capacity. In this context, Suh et al. demonstrated the battery performance of rGO/SnO₂ hybrids synthesized through MW irradiation.¹¹⁸ With a MW treatment of only 10 min, it has been possible to intercalate SnO₂ nanoparticles into the layered structure of graphene. The mesoporous structure of the composite was found to be beneficial for the easy transport of Li⁺ ions. Moreover, the strong interaction between graphene and SnO₂ also reduced the capacity fading during the charge/discharge cycles. Such interaction between the metal oxide and graphene stabilizes the structure of the composite and prevents structural deformation during continuous insertion/deinsertion of Li⁺ ions in the cyclic stability test. Such a stable structure helps to accommodate the volumetric changes during lithiation/delithiation processes. As a result, the anode material exhibited a capacity of 1200 mAh/g at the current density of 50 mA/g. The restacking of graphene sheets has been avoided by the incorporation of SnO₂ nanoparticles, resulting in better rate capability for the electrodes.

Influenced by the good results from previous research, our group evaluated the battery performance of Fe₃O₄-coated graphene materials synthesized through the MW approach.⁶³ In this process, octahedral NCs (ONCs) of Fe₃O₄ were embedded in the rGO sheets, forming a conductive porous carbon network. The defects created by the MW irradiation have been found to be favorable for electron and ion transport. As a result, the composite displayed a specific capacity of 540 mAh/g at the current density of 100 mA/g after 120 cycles. The Fe₃O₄ ONCs-based composite showed better cycling stability than other Fe₃O₄ nanoparticle-based composites. Such result has been attributed to the creation of more defects on the octahedral Fe₃O₄ composite.

In a recent article, Zhang et al. adopted a simple MW-assisted solvothermal approach to fabricate LiB anodes based on graphene and CuC₂O₄·xH₂O.¹¹⁹ The authors demonstrated the enhancement of battery performance, including cycle life, rate capability, and charge/discharge capacity for the hybrid with the addition of graphene. The presence of graphene enhanced the conductive nature of the material, suppressed the volume expansion during cyclic testing, and acted as a buffer to reduce the mechanical stresses. The LiB anode displayed the maximum discharge capacity of 1043 mAh/g after 100 cycles.

In general, for LiBs, Li-based metal oxides are used as the cathode material. However, graphene materials have also been investigated as LiB cathodes in combination with Li-based mixed metal oxides. For example, an advanced composite cathode material has been reported, consisting of LiFePO₄ and graphene.¹²⁰ For the synthesis, the precursor materials were heated in a MW oven for 10 min at a power of 1500 W. The incorporation of nanoparticles shortened the transport pathway of Li⁺ ions and enhanced the diffusion of Li⁺ ions into the pores of the material. Moreover, the nanoparticles enclosed by the conductive graphene sheets increased the stability of the electrode during continuous charge/discharge cycles.

In recent times, secondary batteries based on other types of ions interacting with graphene-based materials have been increasingly investigated. The idea is to use cheaper, more abundant elements, like sodium, sulfur, etc. as replacements for lithium. A MW-processed ternary composite based on SnO₂, graphene and N-doped carbon, synthesized through MW heating, has been reported by Dursun et al.¹²¹ As the anode material for Na-ion batteries, this composite exhibited a maximum capacity of 423 mAh/g after 180 cycles. When compared to the binary SnO₂/graphene, the ternary composite displayed better cycling stability. The authors suggested that the addition of N-doped carbon enhanced the number of defects as well as the electrical conductivity. Apart from Li and Na ion batteries, carbon-based materials have also been investigated for fabricating anodes for lithium–sulfur (Li–S) batteries. For example, a hybrid based on rGO and nanosulfur has recently been fabricated through a sulfur-containing ion-assisted MW approach.¹²² As a cathode material, the composite displayed a maximum reversible capacity of 886 mAh/g at the 0.1 C rate. In this article, the authors clearly demonstrated the enhancement of battery performance through MW treatments. Table 2 shows the performances of different MW-synthesized graphene materials in battery applications.

The main advantages of graphene-based electrodes can be summarized as follows:

1. One of the most commonly used commercial anode materials for LiBs is graphite. However, the specific

Table 2. Battery Performance of Different MW-Synthesized Graphene Materials^a

materials	synthesis method	type of battery	electrolyte	specific capacity	cycling stability	ref
3D graphene/CNT/Ni	MW irradiation	Li-ion battery (anode)	1 M LiPF ₆ in EC/DMC (1:1 v/v)	2395.2 mAh/g at 100 mA/g	648.2 mAh/g after 50 cycles	89
N-doped carbon-coated Si/graphene	MW irradiation	Li-ion battery (anode)	1.3 M LiPF ₆ in 3:7 (v/v) EC and DEC with 10% FEC	1744 mAh/g at 0.1 A/g	662 mAh/g at 1 A/g after 200 cycles	117
rGO/SnO ₂	MW irradiation	Li-ion battery (anode)	1 mol/L LiPF ₆ in EC: DMC (volume ratio = 1:1)	1200 mAh/g at 50 mA/g	84% after 100 cycles	118
graphene/LiFePO ₄	MW heating	Li-ion battery (cathode)	1 M LiPF ₆ in EC and DMC (1:1 v/v)	166.3 mAh/g at 0.1 C	99.5 mAh/g at 0.1 C after 10 cycles	120
N-doped carbon/graphene/SnO ₂	MW irradiation	Na-ion battery (anode)	1 M NaClO ₄ in PC containing 5 wt % FEC	646.5 mAh/g at 1 C	423 mAh/g after 180 cycles	121
rGO/nanosulfur	sulfur containing ion assisted MW irradiation	Li–S battery (cathode)	1 M lithium bis(trifluoromethylsulfonyl) imide and 0.1 M LiNO ₃ in DOL/DME solvent (1:1 v/v)	886 mAh/g at 0.1 C	99.96% after 800 cycles	122
graphene/CNT/Fe	MW irradiation	Li-ion battery (anode)	1 M LiPF ₆ in EC/DMC (1:1 v/v)	2892 mAh/g at mA/g	865 mAh/g after 40 cycles	123
iron oxide-decorated and gelatin-functionalized graphene	MW irradiation	Li-ion battery (anode)	1 M LiPF ₆ in EC/DMC (1:1 v/v)	2518 mAh/g at 100 mA/g	924 mAh/g over 40 cycles	124

^aEC, ethylene carbonate; DEC, diethylene carbonate; FEC, fluoroethylene carbonate; DMC, dimethyl carbonate; DOL, 1,3 dioxolane; DME, 1,2 dimethoxyethane; PC, propylene carbonate; FEC, fluoroethylene carbonate.

capacity of graphite is very limited (372 mAh/g).^{125,126} From the results in Table 2 and from the published data on battery performance, it is clear that graphene-based electrodes exhibit enhanced specific capacity when compared to commercially adopted electrodes.

2. For commercial graphite electrodes, the rates of electron and ion transport are being slowed down through the interlayers of graphite. Such a problem can easily be solved by the utilization of graphene-based electrodes.
3. Enhanced flexibility and high surface area helps the graphene-based batteries to deliver higher charge storage capability than the commercial batteries.
4. The utilization of graphene as an essential component for batteries is advantageous since the addition of graphene improves the electrical conductivity and mechanical stability of the electrodes.

6. CONCLUSIONS AND FUTURE PERSPECTIVES

MW technology has been widely used for the development of porous materials for energy-related applications. In general, carbon materials are found to be excellent MW absorbers, and their oxygenated forms are susceptible to exfoliation and reduction. However, they also play the important additional role of transferring heat to other components of the material, driving chemical reactions and structural changes. One of the most relevant examples is the synthesis of metal oxides/mixed metal oxides from common precursors through MW heating.

In spite of having a few outstanding issues, the MW technology was found to be quite successful for developing carbon-based electrodes for energy storage devices (mainly SCs and secondary batteries). The current Spotlight on Applications summarizes the utilization of MW processes for synthesizing electrodes for energy storage devices, highlighting our endeavors in this area. A brief literature review explained the characteristic features and plausible applications for this “green” synthetic route. After that, the Spotlight on Applications discusses the advantages of MW processing in comparison with other synthetic routes, including hydrothermal, solvothermal, and conventional heating processes. Specifically, we have demonstrated the imperative role of rGO for the synthesis of metal oxides/mixed metal oxides through MW routes. The MW synthetic approach has been found to be advantageous due to the short processing times, high product yield, fast heating rates, the absence of hazardous chemicals, as well as the reliance on simple and inexpensive equipment.

Even after the considerable amount of research already carried out in this field, some vital issues of MW chemistry remain ambiguous, due to the complexity of MW heating. A clear understanding of the MW heating process is highly essential for the wide adoption of the method. The current research is mainly focused on optimization of the MW reaction conditions for specific target applications. Based on the existing challenges, future research work should be focused on the following specific points:

1. Establishment of an accurate mechanism for MW heating is highly essential. In this regard, a more detailed theoretical analysis, including modeling and simulation of the characteristics of MW heating, should be carried out in the near future. Smart control of chemical reactions during MW heating should be explored in order to enhance product yield and quality.

2. With rapid progress in materials chemistry, several in situ characterization techniques have been developed in the past few years. In order to understand the impact of MW heating toward product quality, such in situ techniques should be implemented. Specifically, the precise measurement of temperatures during the MW heating process is still a big challenge. Future research should help to solve this practical problem.
3. Apart from carbon-based materials, the MW technology should also be applicable to other materials. In the specific case of energy storage devices, the MW technology should be extensively developed toward the synthesis and functionalization of other 2D materials like boron nitride (BN), MXenes, etc.
4. For further improvement of the product yield as well as for bulk scale production, appropriate designing of MW reactors should be more emphasized. In order to bring this MW technology from the laboratory to the marketplace, large MW reactors with the capability of industrial-scale production should be designed and implemented in the near future.
5. The MW approach is undoubtedly one of the best time-saving processing methods. However, this is only one among many criteria for full adoption of any synthetic route. Proper compounding, with suitable MW reactors at cheap rates is highly desirable for industrial application. In this respect, it is important to remember that the energy efficiency of MW heating has been found to be much higher than the conventional heating process.
6. MW technology has been shown to be favorable for the fabrication of porous electrodes for SC and batteries; however, such carbon-based electrodes also have some common drawbacks, which need to be resolved soon. For example, in the case of Li-ion batteries, the formation of a solid electrolyte interphase (SEI) during the initial charging cycles is one of the major problems, which tends to decrease the capacity. For SCs, a significant improvement in the energy density should be considered as a key research direction in the future.

We hope that this Spotlight on Applications provides an overview of the current state of MW technology as well as some of the future prospects for its application in the development of advanced, more efficient energy storage materials.

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Notes

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REFERENCES

- (1) Kumar, R.; Sahoo, S.; Joanni, E.; Singh, R. K.; Tan, W. K.; Kar, K. K.; Matsuda, A. Recent Progress in the Synthesis of Graphene and Derived Materials for Next Generation Electrodes of High Performance Lithium Ion Batteries. *Prog. Energy Combust. Sci.* **2019**, *75*, 100786.
- (2) Jiang, S.; Qiao, Y.; Fu, T.; Peng, W.; Yu, T.; Yang, B.; Xia, R.; Gao, M. Integrated Battery–Capacitor Electrodes: Pyridinic N-Doped Porous Carbon-Coated Abundant Oxygen Vacancy Mn–Ni-Layered Double Oxide for Hybrid Supercapacitors. *ACS Appl. Mater. Interfaces* **2021**, *13* (29), 34374–34384.
- (3) Kumar, R.; Sahoo, S.; Joanni, E.; Singh, R. K.; Yadav, R. M.; Verma, R. K.; Singh, D. P.; Tan, W. K.; Perez del Pino, A.; Moshkalev, S. A.; Matsuda, A. A Review on Synthesis of Graphene, H-Bn and MoS₂ for Energy Storage Applications: Recent Progress and Perspectives. *Nano Res.* **2019**, *12* (11), 2655–2694.
- (4) Kumar, R.; Joanni, E.; Singh, R. K.; Singh, D. P.; Moshkalev, S. A. Recent Advances in the Synthesis and Modification of Carbon-Based 2d Materials for Application in Energy Conversion and Storage. *Prog. Energy Combust. Sci.* **2018**, *67*, 115–157.
- (5) Escobar, B.; Martínez-Casillas, D. C.; Pérez-Salcedo, K. Y.; Rosas, D.; Morales, L.; Liao, S. J.; Huang, L. L.; Shi, X. Research Progress on Biomass-Derived Carbon Electrode Materials for Electrochemical Energy Storage and Conversion Technologies. *Int. J. Hydrogen Energy* **2021**, *46* (51), 26053–26073.
- (6) Dubey, P.; Shrivastav, V.; Maheshwari, P. H.; Sundriyal, S. Recent Advances in Biomass Derived Activated Carbon Electrodes for Hybrid Electrochemical Capacitor Applications: Challenges and Opportunities. *Carbon* **2020**, *170*, 1–29.
- (7) Kumar, R.; Singh, R. K.; Singh, D. P.; Joanni, E.; Yadav, R. M.; Moshkalev, S. A. Laser-Assisted Synthesis, Reduction and Micro-Patterning of Graphene: Recent Progress and Applications. *Coord. Chem. Rev.* **2017**, *342*, 34–79.
- (8) Li, Z.; Gadipelli, S.; Li, H.; Howard, C. A.; Brett, D. J. L.; Shearing, P. R.; Guo, Z.; Parkin, I. P.; Li, F. Tuning the Interlayer Spacing of Graphene Laminate Films for Efficient Pore Utilization Towards Compact Capacitive Energy Storage. *Nature Energy* **2020**, *5* (2), 160–168.
- (9) Largeot, C.; Portet, C.; Chmiola, J.; Taberna, P.-L.; Gogotsi, Y.; Simon, P. Relation between the Ion Size and Pore Size for an Electric Double-Layer Capacitor. *J. Am. Chem. Soc.* **2008**, *130* (9), 2730–2731.
- (10) Jiang, X.; Guo, F.; Jia, X.; Zhan, Y.; Zhou, H.; Qian, L. Synthesis of Nitrogen-Doped Hierarchical Porous Carbons from Peanut Shell as a Promising Electrode Material for High-Performance Supercapacitors. *Journal of Energy Storage* **2020**, *30*, 101451.
- (11) Kumar, N.; Salehiyan, R.; Chauke, V.; Joseph Bothhoko, O.; Setshedi, K.; Scriba, M.; Masukume, M.; Sinha Ray, S. Top-Down Synthesis of Graphene: A Comprehensive Review. *FlatChem.* **2021**, *27*, 100224.
- (12) Devi, N.; Sahoo, S.; Kumar, R.; Singh, R. K. A Review of the Microwave-Assisted Synthesis of Carbon Nanomaterials, Metal Oxides/Hydroxides and Their Composites for Energy Storage Applications. *Nanoscale* **2021**, *13* (27), 11679–11711.
- (13) Amini, A.; Latifi, M.; Chaouki, J. Electrification of Materials Processing Via Microwave Irradiation: A Review of Mechanism and Applications. *Appl. Therm. Eng.* **2021**, *193*, 117003.
- (14) Singh, R. K.; Kumar, R.; Singh, D. P.; Savu, R.; Moshkalev, S. A. Progress in Microwave-Assisted Synthesis of Quantum Dots (Graphene/Carbon/Semiconducting) for Bioapplications: A Review. *Materials Today Chemistry* **2019**, *12*, 282–314.
- (15) Gunwant, D.; Vedrtam, A. Microwave Absorbing Properties of Carbon Fiber Based Materials: A Review and Prospective. *J. Alloys Compd.* **2021**, *881*, 160572.
- (16) González, Z.; Acevedo, B.; Predeanu, G.; Axinte, S. M.; Drăgoescu, M.-F.; Slăvescu, V.; Fernandez, J. J.; Granda, M.; Gryglewicz, G.; Melendi-Espina, S. Graphene Materials from Microwave-Derived Carbon Precursors. *Fuel Process. Technol.* **2021**, *217*, 106803.
- (17) Kumar, H.; Sharma, R.; Yadav, A.; Kumari, R. Recent Advancement Made in the Field of Reduced Graphene Oxide-Based Nanocomposites Used in the Energy Storage Devices: A Review. *Journal of Energy Storage* **2021**, *33*, 102032.
- (18) Mensing, J. P.; Lomas, T.; Tuantranont, A. 2d and 3d Printing for Graphene Based Supercapacitors and Batteries: A Review. *Sustainable Materials and Technologies* **2020**, *25*, e00190.
- (19) Wu, X.; Mu, F.; Lin, Z. Three-Dimensional Printing of Graphene-Based Materials and the Application in Energy Storage. *Materials Today Advances* **2021**, *11*, 100157.
- (20) Luo, L.; Wu, H.; Yang, J.; Tang, Z.; Shu, K.; Xu, Y.; Yan, W.; Xu, L. Effects of Microwave Pre-Treatment on the Flotation of Ilmenite and Titanite. *Miner. Eng.* **2020**, *155*, 106452.
- (21) Motasemi, F.; Afzal, M. T. A Review on the Microwave-Assisted Pyrolysis Technique. *Renewable Sustainable Energy Rev.* **2013**, *28*, 317–330.
- (22) Haque, K. E. Microwave Energy for Mineral Treatment Processes—a Brief Review. *Int. J. Miner. Process.* **1999**, *57* (1), 1–24.
- (23) Gao, Y.; Mao, Y.; Song, Z.; Zhao, X.; Sun, J.; Wang, W.; Chen, G.; Chen, S. Efficient Generation of Hydrogen by Two-Step Thermochemical Cycles: Successive Thermal Reduction and Water Splitting Reactions Using Equal-Power Microwave Irradiation and a High Entropy Material. *Appl. Energy* **2020**, *279*, 115777.
- (24) Menéndez, J. A.; Arenillas, A.; Fidalgo, B.; Fernández, Y.; Zubizarreta, L.; Calvo, E. G.; Bermúdez, J. M. Microwave Heating Processes Involving Carbon Materials. *Fuel Process. Technol.* **2010**, *91* (1), 1–8.
- (25) Sonar, M. P.; Rathod, V. K. Microwave Assisted Extraction (MAE) Used as a Tool for Rapid Extraction of Marmelosin from Aegle Marmelos and Evaluations of Total Phenolic and Flavonoids Content, Antioxidant and Anti-Inflammatory Activity. *Chemical Data Collections* **2020**, *30*, 100545.
- (26) Schwenke, A. M.; Hoeppe, S.; Schubert, U. S. Synthesis and Modification of Carbon Nanomaterials Utilizing Microwave Heating. *Adv. Mater.* **2015**, *27* (28), 4113–4141.
- (27) Kim, T.; Jo, C.; Lim, W.-G.; Lee, J.; Lee, J.; Lee, K.-H. Facile Conversion of Activated Carbon to Battery Anode Material Using Microwave Graphitization. *Carbon* **2016**, *104*, 106–111.
- (28) Lei, H.; Wang, Y.; Huo, J. Porous Graphitic Carbon Materials Prepared from Cornstarch with the Assistance of Microwave Irradiation. *Microporous Mesoporous Mater.* **2015**, *210*, 39–45.

- (29) Wang, C.; Ma, D.; Bao, X. Transformation of Biomass into Porous Graphitic Carbon Nanostructures by Microwave Irradiation. *J. Phys. Chem. C* **2008**, *112* (45), 17596–17602.
- (30) Vo, T. K.; Le, V. N.; Quang, D. T.; Song, M.; Kim, D.; Kim, J. Rapid Defect Engineering of UiO-67 (Zr) Via Microwave-Assisted Continuous-Flow Synthesis: Effects of Modulator Species and Concentration on the Toluene Adsorption. *Microporous Mesoporous Mater.* **2020**, *306*, 110405.
- (31) Xu, S.; Zhong, G.; Chen, C.; Zhou, M.; Kline, D. J.; Jacob, R. J.; Xie, H.; He, S.; Huang, Z.; Dai, J.; Brozena, A. H.; Shahbazian-Yassar, R.; Zachariah, M. R.; Anlage, S. M.; Hu, L. Uniform, Scalable, High-Temperature Microwave Shock for Nanoparticle Synthesis through Defect Engineering. *Matter* **2019**, *1* (3), 759–769.
- (32) Wu, X.; Yu, M.; Liu, J.; Li, S.; Zhang, X. sp^3 -Defect and Pore Engineered Carbon Framework for High Energy Density Supercapacitors. *J. Power Sources* **2020**, *464*, 228203.
- (33) Yao, X.; Ke, Y.; Ren, W.; Wang, X.; Xiong, F.; Yang, W.; Qin, M.; Li, Q.; Mai, L. Defect-Rich Soft Carbon Porous Nanosheets for Fast and High-Capacity Sodium-Ion Storage. *Adv. Energy Mater.* **2019**, *9* (6), 1803260.
- (34) Sun, N.; Guan, Z.; Liu, Y.; Cao, Y.; Zhu, Q.; Liu, H.; Wang, Z.; Zhang, P.; Xu, B. Extended “Adsorption–Insertion” Model: A New Insight into the Sodium Storage Mechanism of Hard Carbons. *Adv. Energy Mater.* **2019**, *9* (32), 1901351.
- (35) Sun, Y.; Du, X.; Zhang, J.; Huang, N.; Yang, L.; Sun, X. Microwave-Assisted Preparation and Improvement Mechanism of Carbon Nanotube@NiMn₂O₄ Core-Shell Nanocomposite for High Performance Asymmetric Supercapacitors. *J. Power Sources* **2020**, *473*, 228609.
- (36) Deng, J.-H.; Cheng, L.; Wang, F.-J.; Yu, B.; Li, G.-Z.; Li, D.-J.; Cheng, G.-A. Ultralow Field Emission from Thinned, Open-Ended, and Defected Carbon Nanotubes by Using Microwave Hydrogen Plasma Processing. *Appl. Surf. Sci.* **2015**, *324*, 293–299.
- (37) Liu, W.; Jiang, H.; Zhang, X.; Zhao, Y.; Sun, S.; Qiao, J. Less Defective Graphene Aerogel and Its Application in Microwave-Assisted Biomass Pyrolysis to Prepare H₂-Rich Gas. *J. Mater. Chem. A* **2019**, *7* (48), 27236–27240.
- (38) Lin, W.; Moon, K.-S.; Zhang, S.; Ding, Y.; Shang, J.; Chen, M.; Wong, C.-p. Microwave Makes Carbon Nanotubes Less Defective. *ACS Nano* **2010**, *4* (3), 1716–1722.
- (39) Nguyen, J. J.; Bougher, T. L.; Pour Shahid Saeed Abadi, P.; Sharma, A.; Graham, S.; Cola, B. A. Postgrowth Microwave Treatment to Align Carbon Nanotubes. *J. Micro Nano-Manuf.* **2013**, *1* (1), 014501.
- (40) Kumar, R.; Singh, R. K.; Savu, R.; Dubey, P. K.; Kumar, P.; Moshkalev, S. A. Microwave-Assisted Synthesis of Void-Induced Graphene-Wrapped Nickel Oxide Hybrids for Supercapacitor Applications. *RSC Adv.* **2016**, *6* (32), 26612–26620.
- (41) Kumar, R.; Oh, J.-H.; Kim, H.-J.; Jung, J.-H.; Jung, C.-H.; Hong, W. G.; Kim, H.-J.; Park, J.-Y.; Oh, I.-K. Nanohole-Structured and Palladium-Embedded 3D Porous Graphene for Ultrahigh Hydrogen Storage and CO Oxidation Multifunctionalities. *ACS Nano* **2015**, *9* (7), 7343–7351.
- (42) Kumar, R.; Savu, R.; Singh, R. K.; Joanni, E.; Singh, D. P.; Tiwari, V. S.; Vaz, A. R.; da Silva, E. T. S. G.; Maluta, J. R.; Kubota, L. T.; Moshkalev, S. A. Controlled Density of Defects Assisted Perforated Structure in Reduced Graphene Oxide Nanosheets-Palladium Hybrids for Enhanced Ethanol Electro-Oxidation. *Carbon* **2017**, *117*, 137–146.
- (43) Pei, S.; Cheng, H.-M. The Reduction of Graphene Oxide. *Carbon* **2012**, *50* (9), 3210–3228.
- (44) McAllister, M. J.; Li, J.-L.; Adamson, D. H.; Schniepp, H. C.; Abdala, A. A.; Liu, J.; Herrera-Alonso, M.; Milius, D. L.; Car, R.; Prud'homme, R. K.; Aksay, I. A. Single Sheet Functionalized Graphene by Oxidation and Thermal Expansion of Graphite. *Chem. Mater.* **2007**, *19* (18), 4396–4404.
- (45) Kudin, K. N.; Ozbas, B.; Schniepp, H. C.; Prud'homme, R. K.; Aksay, I. A.; Car, R. Raman Spectra of Graphite Oxide and Functionalized Graphene Sheets. *Nano Lett.* **2008**, *8* (1), 36–41.
- (46) Voiry, D.; Yang, J.; Kupferberg, J.; Fullon, R.; Lee, C.; Jeong, H. Y.; Shin, H. S.; Chhowalla, M. High-Quality Graphene Via Microwave Reduction of Solution-Exfoliated Graphene Oxide. *Science* **2016**, *353* (6306), 1413–1416.
- (47) Ali, M.; Riaz, R.; Anjum, A. S.; Sun, K. C.; Li, H.; Ahn, S.; Jeong, S. H.; Ko, M. J. Microwave-Assisted Ultrafast in-Situ Growth of N-Doped Carbon Quantum Dots on Multiwalled Carbon Nanotubes as an Efficient Electrocatalyst for Photovoltaics. *J. Colloid Interface Sci.* **2021**, *586*, 349–361.
- (48) Kumar, R.; Sahoo, S.; Joanni, E.; Singh, R. K.; Maegawa, K.; Tan, W. K.; Kawamura, G.; Kar, K. K.; Matsuda, A. Heteroatom Doped Graphene Engineering for Energy Storage and Conversion. *Mater. Today* **2020**, *39*, 47–65.
- (49) Bundaleska, N.; Henriques, J.; Abrashev, M.; Botelho do Rego, A. M.; Ferraria, A. M.; Almeida, A.; Dias, F. M.; Valcheva, E.; Arnaudov, B.; Upadhyay, K. K.; Montemor, M. F.; Tatarova, E. Large-Scale Synthesis of Free-Standing N-Doped Graphene Using Microwave Plasma. *Sci. Rep.* **2018**, *8*, 12595.
- (50) Xia, X.; Cheng, C.-F.; Zhu, Y.; Vogt, B. D. Ultrafast Microwave-Assisted Synthesis of Highly Nitrogen-Doped Ordered Mesoporous Carbon. *Microporous Mesoporous Mater.* **2021**, *310*, 110639.
- (51) Skorupska, M.; Ilnicka, A.; Lukaszewicz, J. P. N-Doped Graphene Foam Obtained by Microwave-Assisted Exfoliation of Graphite. *Sci. Rep.* **2021**, *11*, 2044.
- (52) Byeon, H.; Gu, B.; Kim, H.-J.; Lee, J. H.; Seo, I.; Kim, J.; Yang, J. W.; Kim, J.-K. Redox Chemistry of Nitrogen-Doped Cnt-Encapsulated Nitroxide Radical Polymers for High Energy Density and Rate-Capability Organic Batteries. *Chem. Eng. J.* **2021**, *413*, 127402.
- (53) Guo, H.; Ruan, B.; Liu, L.; Zhang, L.; Tao, Z.; Chou, S.; Wang, J.; Liu, H. Capillary-Induced Ge Uniformly Distributed in N-Doped Carbon Nanotubes with Enhanced Li-Storage Performance. *Small* **2017**, *13* (28), 1700920.
- (54) Kumar, R.; Yadav, R. M.; Awasthi, K.; Tiwari, R. S.; Srivastava, O. N. Effect of Nitrogen Variation on the Synthesis of Vertically Aligned Bamboo-Shaped C–N Nanotubes Using Sunflower Oil. *Int. J. Nanosci.* **2011**, *10* (04n05), 809–813.
- (55) Kumar, R.; Yadav, R. M.; Awasthi, K.; Shripathi, T.; Sinha, A. S. K.; Tiwari, R. S.; Srivastava, O. N. Synthesis of Carbon and Carbon–Nitrogen Nanotubes Using Green Precursor: Jatropha-Derived Biodiesel. *J. Exp. Nanosci.* **2013**, *8* (4), 606–620.
- (56) Kumar, R.; Singh, R. K.; Dubey, P. K.; Singh, D. P.; Yadav, R. M.; Tiwari, R. S. Freestanding 3d Graphene–Nickel Encapsulated Nitrogen-Rich Aligned Bamboo Like Carbon Nanotubes for High-Performance Supercapacitors with Robust Cycle Stability. *Adv. Mater. Interfaces* **2015**, *2* (15), 1500191.
- (57) Nasini, U. B.; Bairi, V. G.; Ramasahayam, S. K.; Bourdo, S. E.; Viswanathan, T.; Shaikh, A. U. Phosphorous and Nitrogen Dual Heteroatom Doped Mesoporous Carbon Synthesized Via Microwave Method for Supercapacitor Application. *J. Power Sources* **2014**, *250*, 257–265.
- (58) Umrao, S.; Gupta, T. K.; Kumar, S.; Singh, V. K.; Sultania, M. K.; Jung, J. H.; Oh, I.-K.; Srivastava, A. Microwave-Assisted Synthesis of Boron and Nitrogen Co-Doped Reduced Graphene Oxide for the Protection of Electromagnetic Radiation in Ku-Band. *ACS Appl. Mater. Interfaces* **2015**, *7* (35), 19831–19842.
- (59) Kumar, R.; da Silva, E. T. S. G.; Singh, R. K.; Savu, R.; Alafardov, A. V.; Fonseca, L. C.; Carossi, L. C.; Singh, A.; Khandka, S.; Kar, K. K.; Alves, O. L.; Kubota, L. T.; Moshkalev, S. A. Microwave-Assisted Synthesis of Palladium Nanoparticles Intercalated Nitrogen Doped Reduced Graphene Oxide and Their Electrocatalytic Activity for Direct-Ethanol Fuel Cells. *J. Colloid Interface Sci.* **2018**, *515*, 160–171.
- (60) Kumar, R.; Macedo, W. C.; Singh, R. K.; Tiwari, V. S.; Constantino, C. J. L.; Matsuda, A.; Moshkalev, S. A. Nitrogen–Sulfur Co-Doped Reduced Graphene Oxide-Nickel Oxide Nanoparticle Composites for Electromagnetic Interference Shielding. *ACS Applied Nano Materials* **2019**, *2* (7), 4626–4636.
- (61) Vadahanambi, S.; Jung, J.-H.; Kumar, R.; Kim, H.-J.; Oh, I.-K. An Ionic Liquid-Assisted Method for Splitting Carbon Nanotubes to Produce Graphene Nano-Ribbons by Microwave Radiation. *Carbon* **2013**, *53*, 391–398.

- (62) Kumar, R.; Singh, R. K.; Tiwari, V. S.; Yadav, A.; Savu, R.; Vaz, A. R.; Moshkalev, S. A. Enhanced Magnetic Performance of Iron Oxide Nanoparticles Anchored Pristine/ N-Doped Multi-Walled Carbon Nanotubes by Microwave-Assisted Approach. *J. Alloys Compd.* **2017**, *695*, 1793–1801.
- (63) Kumar, R.; Singh, R. K.; Alaferdov, A. V.; Moshkalev, S. A. Rapid and Controllable Synthesis of Fe₃O₄ Octahedral Nanocrystals Embedded-Reduced Graphene Oxide Using Microwave Irradiation for High Performance Lithium-Ion Batteries. *Electrochim. Acta* **2018**, *281*, 78–87.
- (64) Kumar, R.; Kim, H.-J.; Park, S.; Srivastava, A.; Oh, I.-K. Graphene-Wrapped and Cobalt Oxide-Intercalated Hybrid for Extremely Durable Super-Capacitor with Ultrahigh Energy and Power Densities. *Carbon* **2014**, *79*, 192–202.
- (65) Kumar, R.; Matsuo, R.; Kishida, K.; Abdel-Galeil, M. M.; Suda, Y.; Matsuda, A. Homogeneous Reduced Graphene Oxide Supported NiO-MnO₂ Ternary Hybrids for Electrode Material with Improved Capacitive Performance. *Electrochim. Acta* **2019**, *303*, 246–256.
- (66) Kumar, N. S.; Grekov, D.; Pré, P.; Alappat, B. J. Microwave Mode of Heating in the Preparation of Porous Carbon Materials for Adsorption and Energy Storage Applications – an Overview. *Renewable Sustainable Energy Rev.* **2020**, *124*, 109743.
- (67) Kumar, R.; Abdel-Galeil, M. M.; Ya, K. Z.; Fujita, K.; Tan, W. K.; Matsuda, A. Facile and Fast Microwave-Assisted Formation of Reduced Graphene Oxide-Wrapped Manganese Cobaltite Ternary Hybrids as Improved Supercapacitor Electrode Material. *Appl. Surf. Sci.* **2019**, *481*, 296–306.
- (68) Kumar, R.; Youssry, S. M.; Ya, K. Z.; Tan, W. K.; Kawamura, G.; Matsuda, A. Microwave-Assisted Synthesis of Mn₃O₄-Fe₂O₃/Fe₃O₄@rGO Ternary Hybrids and Electrochemical Performance for Supercapacitor Electrode. *Diamond Relat. Mater.* **2020**, *101*, 107622.
- (69) Yokoyama, H.; Numakura, H.; Koiwa, M. The Solubility and Diffusion of Carbon in Palladium. *Acta Mater.* **1998**, *46* (8), 2823–2830.
- (70) Kumar, R.; Singh, R. K.; Singh, D. P.; Vaz, A. R.; Yadav, R. R.; Rout, C. S.; Moshkalev, S. A. Synthesis of Self-Assembled and Hierarchical Palladium-CNTs-Reduced Graphene Oxide Composites for Enhanced Field Emission Properties. *Mater. Des.* **2017**, *122*, 110–117.
- (71) Hashtroudi, H.; Kumar, R.; Savu, R.; Moshkalev, S.; Kawamura, G.; Matsuda, A.; Shafiei, M. Hydrogen Gas Sensing Properties of Microwave-Assisted 2D Hybrid Pd/rGO: Effect of Temperature, Humidity and UV Illumination. *Int. J. Hydrogen Energy* **2021**, *46* (10), 7653–7665.
- (72) Kumar, R.; Singh, R. K.; Vaz, A. R.; Moshkalev, S. A. Microwave-Assisted Synthesis and Deposition of a Thin ZnO Layer on Microwave-Exfoliated Graphene: Optical and Electrochemical Evaluations. *RSC Adv.* **2015**, *5* (83), 67988–67995.
- (73) Kumar, R.; Youssry, S. M.; Abdel-Galeil, M. M.; Matsuda, A. One-Pot Synthesis of Reduced Graphene Oxide Nanosheets Anchored ZnO Nanoparticles Via Microwave Approach for Electrochemical Performance as Supercapacitor Electrode. *J. Mater. Sci.: Mater. Electron.* **2020**, *31* (18), 15456–15465.
- (74) Kumar, R.; Singh, R. K.; Dubey, P. K.; Singh, D. P.; Yadav, R. M. Self-Assembled Hierarchical Formation of Conjugated 3D Cobalt Oxide Nanobead-Cnt-Graphene Nanostructure Using Microwaves for High-Performance Supercapacitor Electrode. *ACS Appl. Mater. Interfaces* **2015**, *7* (27), 15042–15051.
- (75) Kumar, R.; Singh, R. K.; Singh, A. K.; Vaz, A. R.; Rout, C. S.; Moshkalev, S. A. Facile and Single Step Synthesis of Three Dimensional Reduced Graphene Oxide-NiCoO₂ Composite Using Microwave for Enhanced Electron Field Emission Properties. *Appl. Surf. Sci.* **2017**, *416*, 259–265.
- (76) Kumar, R.; Singh, R. K.; Vaz, A. R.; Savu, R.; Moshkalev, S. A. Self-Assembled and One-Step Synthesis of Interconnected 3D Network of Fe₃O₄/Reduced Graphene Oxide Nanosheets Hybrid for High-Performance Supercapacitor Electrode. *ACS Appl. Mater. Interfaces* **2017**, *9* (10), 8880–8890.
- (77) Kumar, R.; Alaferdov, A. V.; Singh, R. K.; Singh, A. K.; Shah, J.; Kotnala, R. K.; Singh, K.; Suda, Y.; Moshkalev, S. A. Self-Assembled Nanostructures of 3D Hierarchical Faceted-Iron Oxide Containing Vertical Carbon Nanotubes on Reduced Graphene Oxide Hybrids for Enhanced Electromagnetic Interface Shielding. *Composites, Part B* **2019**, *168*, 66–76.
- (78) Kumar, R.; Singh, R. K.; Tiwari, R. S. Growth Analysis and High-Yield Synthesis of Aligned-Stacked Branched Nitrogen-Doped Carbon Nanotubes Using Sesame Oil as a Natural Botanical Hydrocarbon Precursor. *Mater. Des.* **2016**, *94*, 166–175.
- (79) Kumar, R.; Dubey, P. K.; Singh, R. K.; Vaz, A. R.; Moshkalev, S. A. Catalyst-Free Synthesis of a Three-Dimensional Nanoworm-Like Gallium Oxide-Graphene Nanosheet Hybrid Structure with Enhanced Optical Properties. *RSC Adv.* **2016**, *6* (21), 17669–17677.
- (80) Kumar, R.; Singh, R. K.; Singh, D. P.; Savu, R.; Moshkalev, S. A. Microwave Heating Time Dependent Synthesis of Various Dimensional Graphene Oxide Supported Hierarchical ZnO Nanostructures and Its Photoluminescence Studies. *Mater. Des.* **2016**, *111*, 291–300.
- (81) Kumar, R.; Sahoo, S.; Joanni, E.; Singh, R. K.; Tan, W. K.; Kar, K. K.; Matsuda, A. Recent Progress on Carbon-Based Composite Materials for Microwave Electromagnetic Interference Shielding. *Carbon* **2021**, *177*, 304–331.
- (82) Kumar, R.; Singh, R. K.; Vaz, A. R.; Yadav, R. M.; Rout, C. S.; Moshkalev, S. A. Synthesis of Reduced Graphene Oxide Nanosheet-Supported Agglomerated Cobalt Oxide Nanoparticles and Their Enhanced Electron Field Emission Properties. *New J. Chem.* **2017**, *41* (16), 8431–8436.
- (83) Kumar, R.; Dias, W.; Rubira, R. J.; Alaferdov, A. V.; Vaz, A. R.; Singh, R. K.; Teixeira, S. R.; Constantino, C. J.; Moshkalev, S. A. Simple and Fast Approach for Synthesis of Reduced Graphene Oxide-MoS₂ Hybrids for Room Temperature Gas Detection. *IEEE Trans. Electron Devices* **2018**, *65* (9), 3943–3949.
- (84) Kumar, R.; Singh, R. K.; Dubey, P. K.; Kumar, P.; Tiwari, R. S.; Oh, I.-K. Pressure-Dependent Synthesis of High-Quality Few-Layer Graphene by Plasma-Enhanced Arc Discharge and Their Thermal Stability. *J. Nanopart. Res.* **2013**, *15* (9), 1847.
- (85) Miao, C.; Yin, X.; Xia, G.; Zhu, K.; Ye, K.; Wang, Q.; Yan, J.; Cao, D.; Wang, G. Facile Microwave-Assisted Synthesis of Cobalt Diselenide/Reduced Graphene Oxide Composite for High-Performance Supercapacitors. *Appl. Surf. Sci.* **2021**, *543*, 148811.
- (86) Hamra, A.; Lim, H.; Huang, N.; Gowthaman, N.; Nakajima, H.; Rahman, M. M. Microwave Exfoliated Graphene-Based Materials for Flexible Solid-State Supercapacitor. *J. Mol. Struct.* **2020**, *1220*, 128710.
- (87) Wang, W.; Jin, J.; Wu, Y.; Zhang, W.; Jiang, H.; Li, X.; Wang, G. Unique Holey Graphene/Carbon Dots Frameworks by Microwave-Initiated Chain Reduction for High-Performance Compressible Supercapacitors and Reusable Oil/Water Separation. *J. Mater. Chem. A* **2019**, *7* (38), 22054–22062.
- (88) Kim, T.; Chang Kang, H.; Thanh Tung, T.; Don Lee, J.; Kim, H.; Seok Yang, W.; Gyu Yoon, H.; Suh, K. S. Ionic Liquid-Assisted Microwave Reduction of Graphite Oxide for Supercapacitors. *RSC Adv.* **2012**, *2* (23), 8808–8812.
- (89) Bae, S.-H.; Karthikeyan, K.; Lee, Y.-S.; Oh, I.-K. Microwave Self-Assembly of 3d Graphene-Carbon Nanotube-Nickel Nanostructure for High Capacity Anode Material in Lithium Ion Battery. *Carbon* **2013**, *64*, 527–536.
- (90) Sridhar, V.; Kim, H.-J.; Jung, J.-H.; Lee, C.; Park, S.; Oh, I.-K. Defect-Engineered Three-Dimensional Graphene-Nanotube-Palladium Nanostructures with Ultrahigh Capacitance. *ACS Nano* **2012**, *6* (12), 10562–10570.
- (91) Nagaraju, P.; Arivanandhan, M.; Alsalmeh, A.; Alghamdi, A.; Jayavel, R. Enhanced Electrochemical Performance of α -MoO₃/Graphene Nanocomposites Prepared by an in Situ Microwave Irradiation Technique for Energy Storage Applications. *RSC Adv.* **2020**, *10* (38), 22836–22847.
- (92) Vimuna, V. M.; Athira, A. R.; Dinesh Babu, K. V.; Xavier, T. S. Simultaneous Stirring and Microwave Assisted Synthesis of Nanoflakes MnO₂/rGO Composite Electrode Material for Symmetric Super-

capacitor with Enhanced Electrochemical Performance. *Diamond Relat. Mater.* **2020**, *110*, 108129.

(93) Kumar, R.; Youssry, S. M.; Soe, H. M.; Abdel-Galeil, M. M.; Kawamura, G.; Matsuda, A. Honeycomb-Like Open-Edged Reduced-Graphene-Oxide-Enclosed Transition Metal Oxides (NiO/Co₃O₄) as Improved Electrode Materials for High-Performance Supercapacitor. *Journal of Energy Storage* **2020**, *30*, 101539.

(94) Kumar, R.; Youssry, S. M.; Ya, K. Z.; Tan, W. K.; Kawamura, G.; Matsuda, A. Microwave-Assisted Synthesis of Mn₃O₄-Fe₂O₃/Fe₃O₄@rGO Ternary Hybrids and Electrochemical Performance for Supercapacitor Electrode. *Diamond Relat. Mater.* **2020**, *101*, 107622.

(95) Liu, T.; Chai, H.; Jia, D.; Su, Y.; Wang, T.; Zhou, W. Rapid Microwave-Assisted Synthesis of Mesoporous NiMoO₄ Nanorod/Reduced Graphene Oxide Composites for High-Performance Supercapacitors. *Electrochim. Acta* **2015**, *180*, 998–1006.

(96) Yan, K.; Sun, X.; Ying, S.; Cheng, W.; Deng, Y.; Ma, Z.; Zhao, Y.; Wang, X.; Pan, L.; Shi, Y. Ultrafast Microwave Synthesis of Rambutan-Like CMK-3/Carbon Nanotubes Nanocomposites for High-Performance Supercapacitor Electrode Materials. *Sci. Rep.* **2020**, *10*, 6227.

(97) Xu, Z.; Li, Z.; Tan, X.; Holt, C. M.; Zhang, L.; Amirkhiz, B. S.; Mitlin, D. Supercapacitive Carbon Nanotube-Cobalt Molybdate Nanocomposites Prepared Via Solvent-Free Microwave Synthesis. *RSC Adv.* **2012**, *2* (7), 2753–2755.

(98) Sun, Y.; Du, X.; Zhang, J.; Huang, N.; Yang, L.; Sun, X. Microwave-Assisted Preparation and Improvement Mechanism of Carbon Nanotube@ NiMn₂O₄ Core-Shell Nanocomposite for High Performance Asymmetric Supercapacitors. *J. Power Sources* **2020**, *473*, 228609.

(99) Zheng, Y.; Tian, Y.; Liu, S.; Tan, X.; Wang, S.; Guo, Q.; Luo, J.; Li, Z. One-Step Microwave Synthesis of NiO/NiS@ CNT Nanocomposites for High-Cycling-Stability Supercapacitors. *J. Alloys Compd.* **2019**, *806*, 170–179.

(100) Reddy, B. J.; Vickraman, P.; Justin, A. S. Synthesis and Characterization of Graphene/Binary Metal Molybdate (Graphene/Zn_{1-x}Ni_xMoO₄) Nanocomposite for Supercapacitors. *Phys. Status Solidi A* **2019**, *216* (2), 1800595.

(101) Lamiel, C.; Nguyen, V. H.; Roh, C.; Kang, C.; Shim, J.-J. Synthesis of Mesoporous rGO@(Co, Mn)₃O₄ Nanocomposite by Microwave-Assisted Method for Supercapacitor Application. *Electrochim. Acta* **2016**, *210*, 240–250.

(102) Masikhwa, T. M.; Madito, M. J.; Bello, A.; Lekitima, J.; Manyala, N. Microwave-Assisted Synthesis of Cobalt Sulphide Nanoparticle Clusters on Activated Graphene Foam for Electrochemical Supercapacitors. *RSC Adv.* **2017**, *7* (33), 20231–20240.

(103) da Silveira Firmiano, E. G.; Rabelo, A. C.; Dalmascio, C. J.; Pinheiro, A. N.; Pereira, E. C.; Schreiner, W. H.; Leite, E. R. Supercapacitor Electrodes Obtained by Directly Bonding 2D MoS₂ on Reduced Graphene Oxide. *Adv. Energy Mater.* **2014**, *4* (6), 1301380.

(104) Wang, Y.; Zhang, W.; Guo, X.; Liu, Y.; Zheng, Y.; Zhang, M.; Li, R.; Peng, Z.; Zhang, Y.; Zhang, T. One-Step Microwave-Hydrothermal Preparation of NiS/rGO Hybrid for High-Performance Symmetric Solid-State Supercapacitor. *Appl. Surf. Sci.* **2020**, *514*, 146080.

(105) Zhao, F.; Xie, D.; Huang, W.; Song, X.; Aurang Zeb Gul Sial, M.; Wu, H.; Deng, F.; Zhang, Q.; Zou, J.; Zeng, X. Defect-Rich Honeycomb-Like Nickel Cobalt Sulfides on Graphene through Rapid Microwave-Induced Synthesis for Ultrahigh Rate Supercapacitors. *J. Colloid Interface Sci.* **2020**, *580*, 160–170.

(106) Reddy, B. J.; Vickraman, P.; Justin, A. S. Asymmetric Supercapacitor Device Performance Based on Microwave Synthesis of N-Doped Graphene/Nickel Sulfide Nanocomposite. *J. Mater. Sci.* **2019**, *54* (8), 6361–6373.

(107) Ma, T.; Liu, H.; Wang, Y.; Zhang, M. Rapid Construction of Three-Dimensional Sulfur Doped Graphene Supported by NiFeS₂ Interconnected Networks as Convenient Electron/Ion Transport Channels for Flexible Supercapacitors. *Electrochim. Acta* **2019**, *309*, 1–10.

(108) Khamlich, S.; Mokrani, T.; Dhlamini, M.; Mothudi, B.; Maaza, M. Microwave-Assisted Synthesis of Simonkolleite Nanoplatelets on Nickel Foam–Graphene with Enhanced Surface Area for High-

Performance Supercapacitors. *J. Colloid Interface Sci.* **2016**, *461*, 154–161.

(109) Kim, Y.; Cho, E.-s.; Park, S.-J.; Kim, S. One-Pot Microwave-Assisted Synthesis of Reduced Graphene Oxide/Nickel Cobalt Double Hydroxide Composites and Their Electrochemical Behavior. *J. Ind. Eng. Chem.* **2016**, *33*, 108–114.

(110) Fang, J.; Li, M.; Li, Q.; Zhang, W.; Shou, Q.; Liu, F.; Zhang, X.; Cheng, J. Microwave-Assisted Synthesis of Coal-Layered Double Hydroxide/Graphene Oxide Composite and Its Application in Supercapacitors. *Electrochim. Acta* **2012**, *85*, 248–255.

(111) Li, M.; Cheng, J.; Fang, J.; Yang, Y.; Liu, F.; Zhang, X. Nial-Layered Double Hydroxide/Reduced Graphene Oxide Composite: Microwave-Assisted Synthesis and Supercapacitive Properties. *Electrochim. Acta* **2014**, *134*, 309–318.

(112) Sun, D.; Jin, L.; Chen, Y.; Zhang, J.-R.; Zhu, J.-J. Microwave-Assisted in Situ Synthesis of Graphene/Pedot Hybrid and Its Application in Supercapacitors. *ChemPlusChem* **2013**, *78* (3), 227.

(113) Nguyen, V. H.; Shim, J.-J. Ultrasmall SnO₂ Nanoparticle-Intercalated Graphene@ Polyaniline Composites as an Active Electrode Material for Supercapacitors in Different Electrolytes. *Synth. Met.* **2015**, *207*, 110–115.

(114) Sridhar, V.; Park, H. Manganese Nitride Stabilized on Reduced Graphene Oxide Substrate for High Performance Sodium Ion Batteries, Super-Capacitors and Emi Shielding. *J. Alloys Compd.* **2019**, *808*, 151748.

(115) Sarwar, S.; Liu, Z.; Li, J.; Wang, Y.; Wang, R.; Zhang, X. High Performance Hybrid-Capacitor Based on MoTe₂/Graphene through Ultra-Fast, Facile Microwave-Initiated Synthesis. *J. Alloys Compd.* **2020**, *846*, 155886.

(116) Sahoo, S.; Bae, S.-H.; Lee, Y.-S.; Lee, J.-M.; Ahn, J.-M.; Kim, C.-G.; Oh, I.-K. Defect-Engineered Mesoporous Ternary Nanoarchitecture of Zinc-Cobalt-Oxide/Nitrogen-Doped Graphene as Anode Material in Lithium Ion Batteries. *Carbon* **2015**, *94*, 455–463.

(117) Kim, J. M.; Ko, D.; Oh, J.; Lee, J.; Hwang, T.; Jeon, Y.; Hooch Antink, W.; Piao, Y. Electrochemically Exfoliated Graphene as a Novel Microwave Susceptor: The Ultrafast Microwave-Assisted Synthesis of Carbon-Coated Silicon– Graphene Film as a Lithium-Ion Battery Anode. *Nanoscale* **2017**, *9* (40), 15582–15590.

(118) Hoon Suh, D.; Park, S. K.; Nakhanev, P.; Kang, S.-W.; Park, H. S. Microwave Synthesis of SnO₂ Nanocrystals Decorated on the Layer-by-Layer Reduced Graphene Oxide for an Application into Lithium Ion Battery Anode. *J. Alloys Compd.* **2017**, *702*, 636–643.

(119) Zhang, C.; Zheng, B.; Song, Z.; Shi, S.; Mao, H. Microwave-Assisted Synthesis of a Novel CuC₂O₄·xH₂O/Graphene Composite as Anode Material for Lithium Ion Batteries. *Ceram. Int.* **2020**, *46* (1), 1018–1025.

(120) Liu, S.; Yan, P.; Li, H.; Zhang, X.; Sun, W. One-Step Microwave Synthesis of Micro/Nanoscale LiFePO₄/Graphene Cathode with High Performance for Lithium-Ion Batteries. *Front. Chem.* **2020**, *8*, 104.

(121) Dursun, B.; Topac, E.; Alibeyli, R.; Ata, A.; Ozturk, O.; Demir-Cakan, R. Fast Microwave Synthesis of SnO₂@ Graphene/N-Doped Carbons as Anode Materials in Sodium Ion Batteries. *J. Alloys Compd.* **2017**, *728*, 1305–1314.

(122) Lu, Z. W.; Wang, Y. H.; Dai, Z.; Li, X. P.; Zhang, C. Y.; Sun, G. Z.; Gong, C. S.; Pan, X. J.; Lan, W.; Zhou, J. Y.; Xie, E. Q. One-Pot Sulfur-Containing Ion Assisted Microwave Synthesis of Reduced Graphene Oxide@ Nano-Sulfur Fibrous Hybrids for High-Performance Lithium-Sulfur Batteries. *Electrochim. Acta* **2019**, *325*, 134920.

(123) Lee, S.-H.; Sridhar, V.; Jung, J.-H.; Karthikeyan, K.; Lee, Y.-S.; Mukherjee, R.; Koratkar, N.; Oh, I.-K. Graphene–Nanotube–Iron Hierarchical Nanostructure as Lithium Ion Battery Anode. *ACS Nano* **2013**, *7* (5), 4242–4251.

(124) Lee, S.-H.; Kotal, M.; Oh, J.-H.; Sennu, P.; Park, S.-H.; Lee, Y.-S.; Oh, I.-K. Nanohole-Structured, Iron Oxide-Decorated and Gelatin-Functionalized Graphene for High Rate and High Capacity Li-Ion Anode. *Carbon* **2017**, *119*, 355–364.

(125) Lian, P.; Zhu, X.; Xiang, H.; Li, Z.; Yang, W.; Wang, H. Enhanced Cycling Performance of Fe₃O₄–Graphene Nanocomposite

as an Anode Material for Lithium-Ion Batteries. *Electrochim. Acta* **2010**, 56 (2), 834–840.

(126) Zhao, W.; Fang, M.; Wu, F.; Wu, H.; Wang, L.; Chen, G. Preparation of Graphene by Exfoliation of Graphite Using Wet Ball Milling. *J. Mater. Chem.* **2010**, 20 (28), 5817–5819.

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